Western Blot Integrity Atlas

Full reference-proteome molecular-weight audit report with embedded figures and complete tables

Defensive use only. This report helps evaluate whether a claimed western blot result is plausible and what controls are needed. It must not be used to plan, enable, or disguise blot substitution or fabrication.

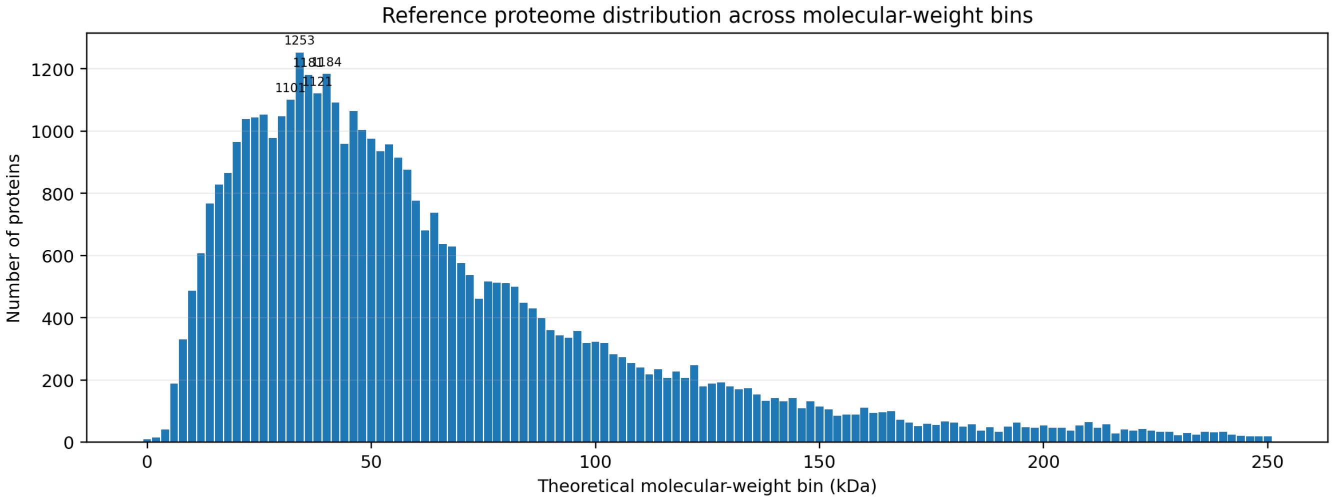
Metric	Value
Proteins parsed	43,712
Molecular-weight bins	323
Bin width	2 kDa
MW range	0.64 to 3895.53 kDa
Median MW	53.36 kDa
Mean MW	72.60 kDa
Proteins <= 50 kDa	20,230 (46.3%)
Proteins 50-100 kDa	14,737 (33.7%)
Proteins > 100 kDa	8,745 (20.0%)
High-priority bins	33
Elevated-priority bins	48
Routine-priority bins	242
Source FASTA	chicken_UP000000539.fasta

Method in one breath

For each protein sequence, the script computes theoretical average molecular weight from residue masses, assigns a molecular-weight bin, applies the external marker, signaling-prefix, and family-pattern annotations, maps supplied public malpractice cases to bins, and calculates an audit-burden priority. Similar apparent molecular weight is never treated as proof of identity; it only increases the need for full-blot provenance and orthogonal controls.

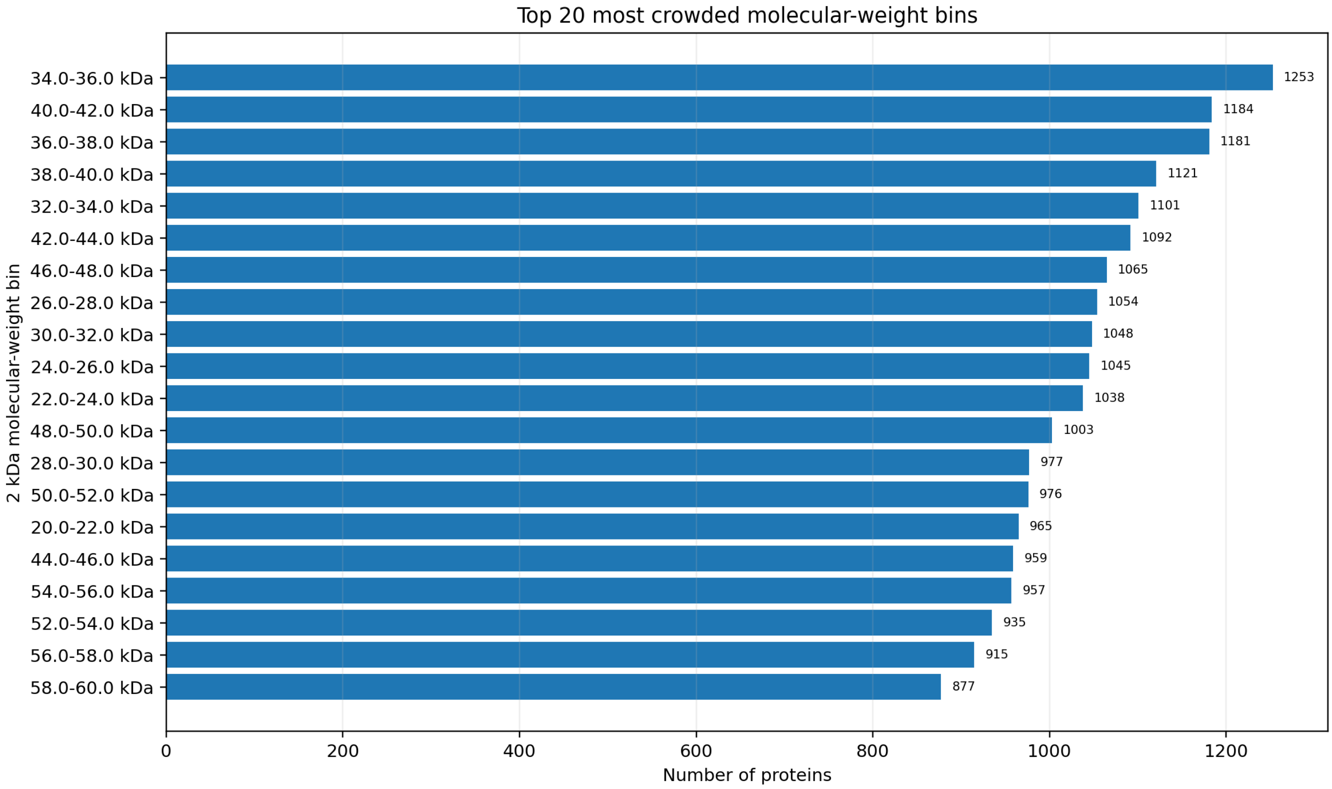
Molecular-weight distribution of the reference proteome.

Proteins were grouped into theoretical molecular-weight bins. Dense gel regions are especially relevant for western blot review because many unrelated proteins occupy nearby molecular-weight neighborhoods.



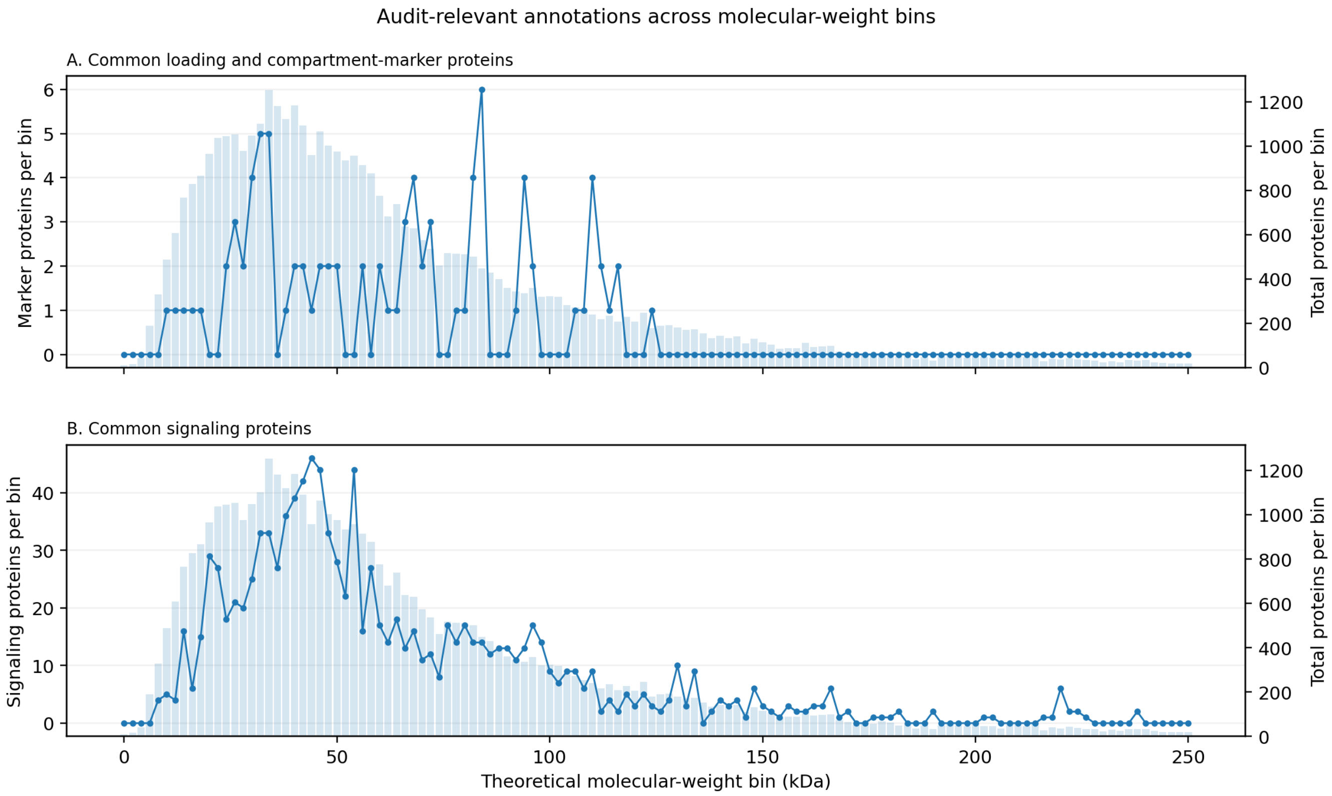
Most crowded molecular-weight neighborhoods.

The top bins are ranked by protein count. These bins are not suspicious by themselves; they indicate where molecular weight alone is a weak identity criterion and where validation documentation becomes more important.



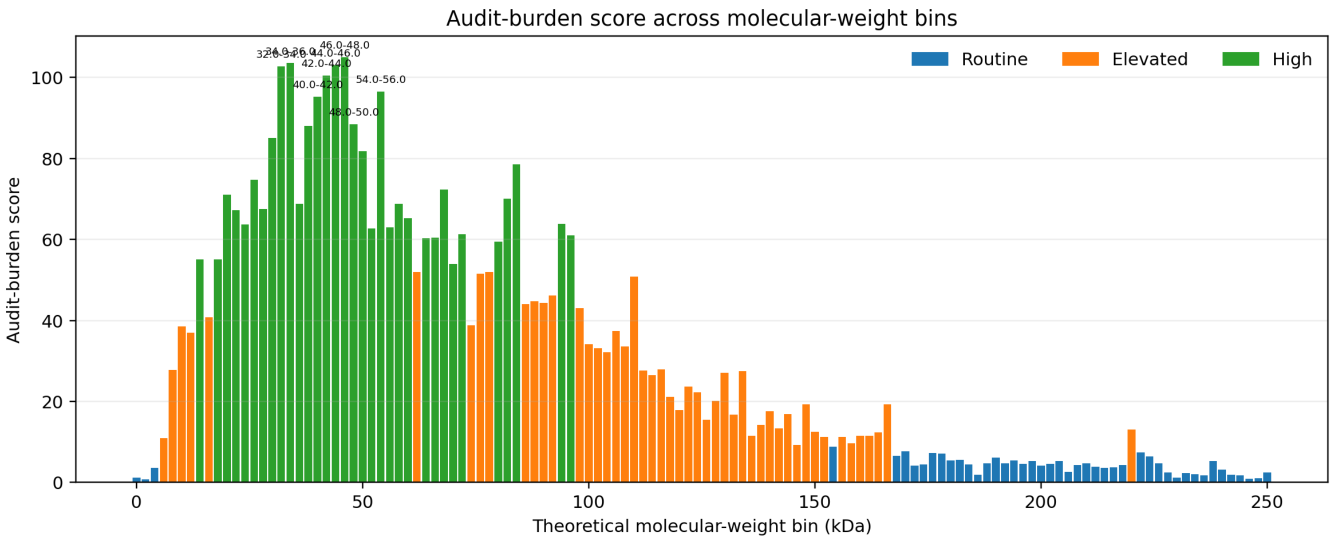
Distribution of common western blot markers and signaling proteins across molecular-weight bins.

The pale background bars show total protein density, while the overlaid lines show audit-relevant annotations. Panel A shows configured loading or compartment markers; Panel B shows configured signaling-prefix matches.



Audit-burden score across molecular-weight bins.

The score integrates protein density, marker count, signaling-prefix count, and configured family/context information. High-scoring bins define high-documentation zones for western blot identity claims.



Highest-audit molecular-weight neighborhoods: score summary

MW bin	N	Markers	Signal	Priority	Score
46.0-48.0 kDa	1065	2	44	high	105.0
34.0-36.0 kDa	1253	5	33	high	103.5
44.0-46.0 kDa	959	1	46	high	103.0
32.0-34.0 kDa	1101	5	33	high	102.8
42.0-44.0 kDa	1092	2	42	high	100.5
54.0-56.0 kDa	957	0	44	high	96.5
40.0-42.0 kDa	1184	2	39	high	95.2
48.0-50.0 kDa	1003	2	33	high	88.5
38.0-40.0 kDa	1121	1	36	high	88.0
30.0-32.0 kDa	1048	4	25	high	85.0
50.0-52.0 kDa	976	2	28	high	81.8
84.0-86.0 kDa	449	6	14	high	78.5
26.0-28.0 kDa	1054	3	21	high	74.8
68.0-70.0 kDa	630	4	16	high	72.2
20.0-22.0 kDa	965	0	29	high	71.0
82.0-84.0 kDa	501	4	14	high	70.0
36.0-38.0 kDa	1181	0	27	high	68.8
58.0-60.0 kDa	877	0	27	high	68.8
28.0-30.0 kDa	977	2	20	high	67.5
22.0-24.0 kDa	1038	0	27	high	67.2
60.0-62.0 kDa	776	2	17	high	65.2
94.0-96.0 kDa	337	4	13	high	63.9
24.0-26.0 kDa	1045	2	18	high	63.8
56.0-58.0 kDa	915	2	16	high	63.0
52.0-54.0 kDa	935	0	22	high	62.8

Highest-audit molecular-weight neighborhoods

These bins combine protein density with configured markers, signaling-prefix matches, and family-rich regions. They are not substitution recommendations; they are places where reviewers should be most insistent about source files and controls.

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
46.0-48.0 kDa	1065	2	44	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	NTSR1, PDLIM7, PHYHIPL [Actin family], TMEM237, IDH2, UBXN6, LIMS1, LUC7L2, GLA, PAX6, SPOCK1, MAP2K5 [MAPK cascade], SS18, RSL1D1, SEC61A2, SCRN1, SMYD2 [Histone family], ABRAXAS1
34.0-36.0 kDa	1253	5	33	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	UNG, STUB1, MADPRT, SLC35E3, GSG1, RASSF3, ORC6, SYNDIG1, CD72L1, CCDC117, , ZNF277, RAX, CCNDBP1 [Cell-cycle CDK/cyclin], MTCH1 [Mitochondrial markers], SPDEF, SPARC, CD200R1A
44.0-46.0 kDa	959	1	46	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	LAMP3, BUB3, B3GNT7, DCP2, PIPOX, MAPK9 [MAPK cascade], KLHL31, RPRD1B, SERPINE3, SAMD14, MYZAP, CDC37, SEH1L, NT5C3B, SERPINE2, OXTR, RHBDL3, TRAPPC13
32.0-34.0 kDa	1101	5	33	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	MBP, SPAG16, THAP8, PHYHD1, PSMD14, KIAA1191, , FAM228, , MXD1, RALY, NECAP1, MTCH2 [Mitochondrial markers], HUS1, , GRAMD2, RSPO3, AQP3
42.0-44.0 kDa	1092	2	42	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; p53 pathway	high	MEIS1, PRSS23, TMEM164, ATG4A, UBE3D, DEK, BGV, LOC100858953, ACTA2 [Actin family], ZDHHC2, GORAB [Actin family], ACTBL2 [Actin family], NR1I3, STK17B, B3GNT3, ACTC1 [Actin family], ERCC8, DPEP1
54.0-56.0 kDa	957	0	44	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	TRMT13, SPNS3, PHGDH, SF3A3, IL17REL, HSPA13 [Heat shock proteins], SYT17, PLEKHD1, THUMPD2, HSPA14 [Heat shock proteins], SLC38A7, TUBE1 [Tubulin family], MGAT4F, MATN1, GTPBP3 [Mitochondrial markers], CCDC162, ENTPD8, MFSD5
40.0-42.0 kDa	1184	2	39	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	MITF, PARP11, RBFOX2, PDGFRL, MEIS2, CPPED1, , SH3GL3, DMTF1 [Cell-cycle CDK/cyclin], , CFAP410, IL10RB, FAM45A, USP41, SH3GL2, ALS2, RPLP0, FGFR1OP
48.0-50.0 kDa	1003	2	33	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family	high	SEPTIN6, U2SURP, KMT5A [Histone family], FADS1, WRAP73, RSRC2, STK24, GLUL, TTLL9 [Tubulin family], MTHFSD, KCTD18, SLC16A7, BZW1, DLST, DRD3, SYT13, RARB, PDE7B
38.0-40.0 kDa	1121	1	36	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	HSD11B1L, INSYN2B2, GALM, MAP2K6 [MAPK cascade], , NODAL, MAFK, SNX20, LDB2, SDF4, ALDOC, MAP2K6 [MAPK cascade], ELAVL1, RNF215, SLC35C2, LOC107050717, TCF7L2, B4GALT4
30.0-32.0 kDa	1048	4	25	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Lamin family; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	TMEM241, , ASB13, ITM2B, TMEM220, DENND1A, MLEC, DRAM2, ATG5, HOXD12, TFAM [Mitochondrial markers], ELOVL3, SNAI2, RALB, ZFP511, AGPAT2, , LOC101747821
50.0-52.0 kDa	976	2	28	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	TUBB6 [Tubulin family], SMARCD2, SMOC1, TRMU [Mitochondrial markers], FKBP10, BTBD10, RAD18, AZIN1, CYP2AB5, DLST [Mitochondrial markers], AS3MT, SHANK2, MAP9, STYK1, SLC29A3, CHRNA10, WIPI2 [Actin family], LOC121106919
84.0-86.0 kDa	449	6	14	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers; p53 pathway	high	EVI5L, APBB2, MTSS1L, SGIP1 [Actin family], GTPBP4, TLE3, GPAT2 [Mitochondrial markers], IL4R, NCKIPSD [Actin family], JPH3, COG4, MARCHF7, HSP90AA1 [Heat shock proteins], ELMO1, KIF2A, LRFN5, LRRC1, RUBCNL
26.0-28.0 kDa	1054	3	21	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	RABL3, PPP1R1B, NKIRAS1 [Actin family], BATF, TSN, DHRS11, C1QB, BCL7B [Apoptosis caspase/BCL], DNAJB9, LOC121106941, TPPP [Tubulin family], NFU1, PPM1L, , , COQ10A, PPP1R2, COQ10B [Mitochondrial markers]
68.0-70.0 kDa	630	4	16	Actin family; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	SEPTIN9, RPA1, AIFM3, KLHL3, PTPN9, LOC419409, SLC18A2, GUCY1B3, , KLHL20, BFSP1, LOC101751434, ZNF606, LOC419429, SCAI, COLGALT1, SLC25A12, CHRDL2
20.0-22.0 kDa	965	0	29	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; p53 pathway	high	CBFB, , JDP2, RAB22A, GREM2, FSBP, TMX1, ALG14, NTPCR, FAM180A, FAM168B, CLCF1, CRYAB, CGA, RPP25L, SAMD12, , LOC121109670
82.0-84.0 kDa	501	4	14	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	CEP63, ANKRD13D, VIT, PCASP2 [Apoptosis caspase/BCL], VRTN, TRAK1, ARNT, SMYD4, USP49, IL23R, TLE4, FGFR4, CBARP, SLC26A6, RPS6KA2, BRD2, MARCHF7, MAPT
36.0-38.0 kDa	1181	0	27	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	PNKD, CHIR3B3, LOC112533538, MTERF4 [Mitochondrial markers], SUSD4, AASDHPPT, TMEM189, USF1, CCN3 [Cell-cycle CDK/cyclin], IRF1, CD274, PPP1R3C, LUZP2, TOB1, LOC121112382, APTX, RP2, RCN1
58.0-60.0 kDa	877	0	27	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	GJA10, LACTB, PKM, KPNA2, LACTB, LIPI, SOAT1, PLXDC2, CAMK2B, DPYS, RFX5, CTTN [Actin family], HEXA, , SESN3, PFKFB1, CCDC77, TCF12
28.0-30.0 kDa	977	2	20	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	RCHY1, KIAA1644, LOC421419, UTP23, FOLR3, PTPRF, RTRAF, APOA1, RPL8, , TPRG1L, ZCCHC9, LDLRAD4, THYN1, , PEX11B, SHISA9, NQO1
22.0-24.0 kDa	1038	0	27	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; Tubulin family; p53 pathway	high	GATD3AL1, , ORMDL3, VSTM2L, [Histone family], IL15, PLCXD1, LOC101751545 [Cell-cycle CDK/cyclin], CRYBG3, CLEC3A, GUCA1B, TRNP1, UBE2T, UBE2K, CAMKK1, DTD2, ANKRD22, CLDN19
60.0-62.0 kDa	776	2	17	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	YRK, UGT1A1, ZFYVE28B, ATE1, , SRC, ASIC1, ESAM, C8B, LOC107050229, GPT2, LOC107050463, CAT, DPYS, FZD5, CABIN1, PLD5, PIGQ
94.0-96.0 kDa	337	4	13	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; p53 pathway	high	ABR, AGBL2, HLCS, IKBKE [NF-kB/Rel], SLMAP, GPAM [Mitochondrial markers], LOC430766, SOBP, LRP3, STOX1, CCSER2, BACH2, ZC3H12B, ST14, MOBP, DRC7, PRLR [Actin family], ESYT3
24.0-26.0 kDa	1045	2	18	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; Tubulin family; p53 pathway	high	, PGLS, ZFAND6, TCTEX1D4, EIF4H, PSMD10, FEV, NUDT16L1, CHMP2B, NKAIN3 [Actin family], MPZL2, TIMP4, AKAP14, IL2RA, HPRT1, CCDC59, GSTK1, LIN54
56.0-58.0 kDa	915	2	16	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family	high	FTO, LOC107056139, ZBTB8A, KPNA3, TMEM39B, HCK, SH2D2A, HEPACAM2, CYP2AC1, TH, ALDH3B1, , LRRC71, ZNF606, DUS1L, TBLX1, CHRNB2, ENTPD8
52.0-54.0 kDa	935	0	22	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	POLL, PRR33, CRBN, RGS6, PPP2R2A, MRPS27, CPQ, CCDC15, KATNA1, LOC121108224, PTGER4, TCTN3, PPP2R2D, SCAI, GLRA2, DCTN4 [Actin family], RGS6, PARK2
72.0-74.0 kDa	537	3	12	Actin family; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	GRAMD1C, TMEM62, RHOBTB1, RUFY1, NUMB, SLC6A14, HSPA5 [Heat shock proteins], MTMR6, RASGRP3, PRMT5, FRMD1, TP63, FOXP1, CAPN13, LDB3, CHFR, TESK2, RAD21
96.0-98.0 kDa	359	2	17	Actin family; Apoptosis caspase/BCL; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	VAV3, RIN2, PLA2G4EL6, EXOC6B, DDR2L, ZNF438, BBX, , EPB41L1, AHR2, VWA2, SIK2, POSTN, PRKD1, EXOC6, HECTD3, BCAR3, FGD4
66.0-68.0 kDa	637	3	13	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel	high	RAP1GDS1, SACM1L, BFSP2, NTN4L, TDP1, KLHL10, RNGTT, SLC6A2, RPAP2, ENC1, IL1RAP, PTCHD3, KBTBD2, FBXW7, P2RX7, SLC2A1, FBXO21, SLC6A20
64.0-66.0 kDa	739	1	18	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; p53 pathway	high	KPNA5, ALB, KIZ, GPC5, ENTPD1, HABP2, IL2RB, GALNT18, INSC, TIMD4, UBQLN4, CNOT6, WFIKKN2, ENO4, ALAS1, CCDC88B, CCAR1 [Apoptosis caspase/BCL], LOC107056275
80.0-82.0 kDa	511	1	17	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	SLC7A4, SNAP91, MTA1, VIT, MARK1, LOC419333, , AKAP8L, MRE11, SSRP1, RTN1, MME, TRIM9, SEMA3B, LEKR1, SEC23B, RASEF, PFKP

Common loading/compartment markers present in this proteome file

MW kDa	Gene	Accession	Entry	Protein	MW bin
10.36	VCL	A0A8V0ZI76	A0A8V0ZI76_CHICK	Vinculin	10.0-12.0 kDa
13.58	H2AFZ	Q5ZMD6	H2AZ_CHICK	Histone H2A.Z	12.0-14.0 kDa
15.25	TOMM20	A0A8V0Y6E3	A0A8V0Y6E3_CHICK	Translocase of outer mitochondrial membrane 20	14.0-16.0 kDa
16.42	TOMM20	A0A1D5PKT3	A0A1D5PKT3_CHICK	Translocase of outer mitochondrial membrane 20	16.0-18.0 kDa
19.33	GAPDH	A0A8V0ZG53	A0A8V0ZG53_CHICK	glyceraldehyde-3-phosphate dehydrogenase (phosphorylating)	18.0-20.0 kDa
24.72	GAPDH	A0A8V0ZN93	A0A8V0ZN93_CHICK	glyceraldehyde-3-phosphate dehydrogenase (phosphorylating)	24.0-26.0 kDa
25.97	PCNA	A0A8V0Z778	A0A8V0Z778_CHICK	DNA sliding clamp PCNA	24.0-26.0 kDa
26.66	GAPDH	A0A8V0ZLG9	A0A8V0ZLG9_CHICK	glyceraldehyde-3-phosphate dehydrogenase (phosphorylating)	26.0-28.0 kDa
27.54	PCNA	A0A8V0ZEF6	A0A8V0ZEF6_CHICK	DNA sliding clamp PCNA	26.0-28.0 kDa
27.63	COX4I1	A0A8V0XW68	A0A8V0XW68_CHICK	Cytochrome c oxidase subunit 4	26.0-28.0 kDa
28.45	VDAC2	A0A8V0ZHC7	A0A8V0ZHC7_CHICK	Non-selective voltage-gated ion channel VDAC2	28.0-30.0 kDa
28.89	PCNA	Q9DEA3	PCNA_CHICK	DNA sliding clamp PCNA	28.0-30.0 kDa
30.60	VDAC1	A0A8V0YU39	A0A8V0YU39_CHICK	Non-selective voltage-gated ion channel VDAC1	30.0-32.0 kDa
30.71	VDAC1	E1BYN7	E1BYN7_CHICK	Non-selective voltage-gated ion channel VDAC1	30.0-32.0 kDa
31.80	VDAC2	A0A8V0ZMQ5	A0A8V0ZMQ5_CHICK	Non-selective voltage-gated ion channel VDAC2	30.0-32.0 kDa
31.82	TBP	A0A8V0Y299	A0A8V0Y299_CHICK	TATA-box binding protein	30.0-32.0 kDa
32.61	VDAC1	A0A8V0Z0Y0	A0A8V0Z0Y0_CHICK	Non-selective voltage-gated ion channel VDAC1	32.0-34.0 kDa
33.13	TBP	O13270	TBP_CHICK	TATA-box-binding protein	32.0-34.0 kDa
33.18	TBP	Q6JLB1	Q6JLB1_CHICK	TATA-box binding protein	32.0-34.0 kDa
33.70	VDAC2	A0A8V0ZV58	A0A8V0ZV58_CHICK	Non-selective voltage-gated ion channel VDAC2	32.0-34.0 kDa
33.80	VDAC2	A0A8V0ZHE8	A0A8V0ZHE8_CHICK	Non-selective voltage-gated ion channel VDAC2	32.0-34.0 kDa
34.29	RPLP0	P47826	RLA0_CHICK	Large ribosomal subunit protein uL10	34.0-36.0 kDa
34.56	VDAC2	A0A8V0ZHE3	A0A8V0ZHE3_CHICK	Non-selective voltage-gated ion channel VDAC2	34.0-36.0 kDa
34.66	VDAC2	A0A8V0ZPI5	A0A8V0ZPI5_CHICK	Non-selective voltage-gated ion channel VDAC2	34.0-36.0 kDa
35.69	VDAC2	A0A8V0ZPZ5	A0A8V0ZPZ5_CHICK	Non-selective voltage-gated ion channel VDAC2	34.0-36.0 kDa
35.70	GAPDH	P00356	G3P_CHICK	Glyceraldehyde-3-phosphate dehydrogenase	34.0-36.0 kDa
39.32	RPLP0	A0A8V0ZJP1	A0A8V0ZJP1_CHICK	Large ribosomal subunit protein uL10	38.0-40.0 kDa
40.02	RPLP0	A0A8V0ZQJ3	A0A8V0ZQJ3_CHICK	Large ribosomal subunit protein uL10	40.0-42.0 kDa
41.74	ACTB	P60706	ACTB_CHICK	Actin, cytoplasmic 1	40.0-42.0 kDa
42.04	ACTB	A0A8V0ZIN3	A0A8V0ZIN3_CHICK	Actin, beta	42.0-44.0 kDa
43.05	ACTB	A0A8V0ZIN7	A0A8V0ZIN7_CHICK	Actin, beta	42.0-44.0 kDa
44.11	ACTB	A0A8V0ZH84	A0A8V0ZH84_CHICK	Actin, beta	44.0-46.0 kDa
46.48	TUBB2A	A0A8V0YNJ8	A0A8V0YNJ8_CHICK	Tubulin beta chain	46.0-48.0 kDa
47.62	TUBB2A	A0A8V0YNJ4	A0A8V0YNJ4_CHICK	Tubulin beta chain	46.0-48.0 kDa
48.33	CALR	A0A8V0X980	A0A8V0X980_CHICK	Calreticulin	48.0-50.0 kDa
48.60	TUBB2B	A0A8V0YCW1	A0A8V0YCW1_CHICK	Tubulin beta chain	48.0-50.0 kDa
50.14	TUBA1A	P02552	TBA1A_CHICK	Tubulin alpha-1A chain	50.0-52.0 kDa
50.41	TUBB3	F1NMU4	F1NMU4_CHICK	Tubulin beta chain	50.0-52.0 kDa
57.41	P4HB	P09102	PDIA1_CHICK	Protein disulfide-isomerase	56.0-58.0 kDa
57.63	P4HB	A0A8V1AL50	A0A8V1AL50_CHICK	Protein disulfide-isomerase	56.0-58.0 kDa
61.20	LMNB1	A0A8V0YIA8	A0A8V0YIA8_CHICK	Lamin B1	60.0-62.0 kDa
61.55	LMNB1	A0A8V0YN37	A0A8V0YN37_CHICK	Lamin B1	60.0-62.0 kDa
63.64	HSPA8	A0A8V1A8J4	A0A8V1A8J4_CHICK	Heat shock protein family A (Hsp70) member 8	62.0-64.0 kDa
65.96	CANX	A0A8V0YS93	A0A8V0YS93_CHICK	Calnexin	64.0-66.0 kDa
66.53	LMNB1	P14731	LMNB1_CHICK	Lamin-B1	66.0-68.0 kDa
66.55	LMNB1	A0A8V0YN43	A0A8V0YN43_CHICK	Lamin-B1	66.0-68.0 kDa
67.46	EEF2	A0A8V1AD37	A0A8V1AD37_CHICK	Elongation factor 2	66.0-68.0 kDa
68.13	CANX	A0A8V0YS98	A0A8V0YS98_CHICK	Calnexin	68.0-70.0 kDa
68.64	SDHA	A0A8V0Y0D8	A0A8V0Y0D8_CHICK	succinate dehydrogenase	68.0-70.0 kDa
68.96	HSPA8	A0A8V1A5Z3	A0A8V1A5Z3_CHICK	Heat shock protein family A (Hsp70) member 8	68.0-70.0 kDa
69.37	HSPA8	A0A8V1AB30	A0A8V1AB30_CHICK	Heat shock protein family A (Hsp70) member 8	68.0-70.0 kDa
70.33	HSPA8	A0A1D5PFJ6	A0A1D5PFJ6_CHICK	Heat shock protein family A (Hsp70) member 8	70.0-72.0 kDa
70.83	HSPA8	O73885	HSP7C_CHICK	Heat shock cognate 71 kDa protein	70.0-72.0 kDa
72.93	SDHA	Q9YHT1	SDHA_CHICK	Succinate dehydrogenase [ubiquinone] flavoprotein subunit, mitochondrial	72.0-74.0 kDa
72.94	SDHA	A0A8V0XRG9	A0A8V0XRG9_CHICK	Succinate dehydrogenase [ubiquinone] flavoprotein subunit, mitochondrial	72.0-74.0 kDa
73.16	LMNA	P13648	LMNA_CHICK	Lamin-A	72.0-74.0 kDa
78.05	HSPA8	A0A8V1A5Y2	A0A8V1A5Y2_CHICK	Heat shock protein family A (Hsp70) member 8	78.0-80.0 kDa
81.90	HSP90AA1	A0A8V0YRK7	A0A8V0YRK7_CHICK	Heat shock protein 90 alpha family class A member 1	80.0-82.0 kDa
82.24	TFRC	A0A8V0YQW3	A0A8V0YQW3_CHICK	Transferrin receptor protein 1	82.0-84.0 kDa
82.55	HSP90AB1	A0A8V0ZAR1	A0A8V0ZAR1_CHICK	Heat shock protein 90 alpha family class B member 1	82.0-84.0 kDa
82.63	TUBA1A	A0A8V0ZZM1	A0A8V0ZZM1_CHICK	Tubulin alpha chain-like 3	82.0-84.0 kDa
83.43	HSP90AB1	Q04619	HS90B_CHICK	Heat shock cognate protein HSP 90-beta	82.0-84.0 kDa
84.06	HSP90AA1	P11501	HS90A_CHICK	Heat shock protein HSP 90-alpha	84.0-86.0 kDa
84.12	HSP90AA1	A0A1D5PUL6	A0A1D5PUL6_CHICK	Heat shock protein 90 alpha family class A member 1	84.0-86.0 kDa
84.55	HSP90AA1	A0A8V0Z557	A0A8V0Z557_CHICK	Heat shock protein 90 alpha family class A member 1	84.0-86.0 kDa
84.82	TFRC	A0A8V0YQV2	A0A8V0YQV2_CHICK	Transferrin receptor protein 1	84.0-86.0 kDa
85.64	TFRC	A0A8V0YQT2	A0A8V0YQT2_CHICK	Transferrin receptor protein 1	84.0-86.0 kDa
85.66	TFRC	Q90997	TFR1_CHICK	Transferrin receptor protein 1	84.0-86.0 kDa
92.65	EEF2	A0A8V1AFQ0	A0A8V1AFQ0_CHICK	Elongation factor 2	92.0-94.0 kDa
95.36	EEF2	A0A1D5PS29	A0A1D5PS29_CHICK	Elongation factor 2	94.0-96.0 kDa
95.38	EEF2	Q90705	EF2_CHICK	Elongation factor 2	94.0-96.0 kDa
95.44	EEF2	A0A8V1AHL6	A0A8V1AHL6_CHICK	Elongation factor 2	94.0-96.0 kDa
95.61	TFRC	A0A8V0YSJ4	A0A8V0YSJ4_CHICK	Transferrin receptor protein 1	94.0-96.0 kDa
96.53	TFRC	A0A8V0YJE5	A0A8V0YJE5_CHICK	Transferrin receptor protein 1	96.0-98.0 kDa
97.40	EEF2	A0A8V1A5F0	A0A8V1A5F0_CHICK	Elongation factor 2	96.0-98.0 kDa

MW kDa	Gene	Accession	Entry	Protein	MW bin
107.56	ATP1A1	A0A8V0ZX92	A0A8V0ZX92_CHICK	Sodium/potassium-transporting ATPase subunit alpha	106.0-108.0 kDa
108.71	ATP1A1	A0A8V0ZX55	A0A8V0ZX55_CHICK	Sodium/potassium-transporting ATPase subunit alpha	108.0-110.0 kDa
110.02	ATP1A1	A0A8V1A771	A0A8V1A771_CHICK	Sodium/potassium-transporting ATPase subunit alpha	110.0-112.0 kDa
110.78	ATP1A1	A0A8V1A2H4	A0A8V1A2H4_CHICK	Sodium/potassium-transporting ATPase subunit alpha	110.0-112.0 kDa
111.28	ATP1A1	A0A8V1A755	A0A8V1A755_CHICK	Sodium/potassium-transporting ATPase subunit alpha	110.0-112.0 kDa
111.81	VCL	A0A8V0ZQZ7	A0A8V0ZQZ7_CHICK	Vinculin	110.0-112.0 kDa
112.23	ATP1A1	P09572	AT1A1_CHICK	Sodium/potassium-transporting ATPase subunit alpha-1	112.0-114.0 kDa
112.80	ATP1A1	A0A8V1A4S9	A0A8V1A4S9_CHICK	Sodium/potassium-transporting ATPase subunit alpha	112.0-114.0 kDa
114.37	ATP1A1	A0A8V1A4Q7	A0A8V1A4Q7_CHICK	Sodium/potassium-transporting ATPase subunit alpha	114.0-116.0 kDa
116.32	ATP1A1	A0A8V1A740	A0A8V1A740_CHICK	Sodium/potassium-transporting ATPase subunit alpha	116.0-118.0 kDa
117.13	VCL	A0A8V0ZI80	A0A8V0ZI80_CHICK	Vinculin	116.0-118.0 kDa
124.56	VCL	P12003	VINC_CHICK	Vinculin	124.0-126.0 kDa

Complete molecular-weight bin summary, low to high

All molecular-weight bins are shown. Long Families and Examples / audit families cells wrap and continue across pages.

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
0.0-2.0 kDa	10	0	0	Histone family	routine	, , AGT, GNRH2, [Histone family], , ADA2, , NTS, ME1
2.0-4.0 kDa	16	0	0		routine	NPPC, MLN, EIF5A2, , SCT, AMPD1, GAL, RPN1, , HPX, , CLPS, UBLCP1, SLN, BLACAT1, RPN2
4.0-6.0 kDa	42	0	0	Heat shock proteins; Mitochondrial markers	routine	C12orf75, , CRABP2, PYY, , USP49, , , , GLI2, , , , , , , TMSB4B,
6.0-8.0 kDa	189	0	0	Heat shock proteins; Mitochondrial markers	elevated	, , , PLN, PLN, TOMM5 [Mitochondrial markers], , , , COPS9, , C14orf2 [Mitochondrial markers], LOC107055098, , MT3, , , KCTD9
8.0-10.0 kDa	330	0	4	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Mitochondrial markers; Tubulin family; p53 pathway	elevated	TMEM167A, , PKIA, , , GNG13, , , LOC100859314, GNG12, UBA52, SYNE2, ADM, DEFB4A, , [Histone family], ,
10.0-12.0 kDa	488	1	5	Actin family; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; p53 pathway	elevated	LOC418813, LOC121110032, COX7A2 [Mitochondrial markers], SEC61B, , , , LOC100858797, , IFI27L2, VAPB, LOC426913, TCF4, ZPBP1, , IMMP2L [Mitochondrial markers], , GYPC
12.0-14.0 kDa	608	1	4	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family	elevated	, DDTL, LOC431325, SDHAF4 [Mitochondrial markers], TSC22D2, , PTMA, HNRNPK, CHTF8, LOC107055128, UFM1, , LYRM7, IKZF5, IKZF2, , LOC121107576, EMC6
14.0-16.0 kDa	767	1	16	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	PCBD2, FAM103A1, RPL35A, CYCS [Mitochondrial markers], IMMP2L [Mitochondrial markers], H2A-IX [Histone family], SRP14, IMMP2L [Mitochondrial markers], , MTPN, , LSM7, LOC121107546, NDUFS6 [Mitochondrial markers], , GABARAPL1, ,
16.0-18.0 kDa	828	1	6	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; p53 pathway	elevated	FABP7, PTPN7, , LITAF, , , , PNRC2, , LOC396480, , CASK, , UMAD1, MCFD2, RBM3, LOC121108846, LIN52
18.0-20.0 kDa	866	1	15	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	RAMP3, SCT, MMGT1, UBE2I, GCSH [Mitochondrial markers], NDUFAF3, ZCCHC10, C9orf85, NEU3, DR1, , , SFT2D1, GTSF1, , CPLX4, HRASLS, PPP1R17
20.0-22.0 kDa	965	0	29	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; p53 pathway	high	CBFB, , JDP2, RAB22A, GREM2, FSBP, TMX1, ALG14, NTPCR, FAM180A, FAM168B, CLCF1, CRYAB, CGA, RPP25L, SAMD12, , LOC121109670
22.0-24.0 kDa	1038	0	27	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; Tubulin family; p53 pathway	high	GATD3AL1, , ORMDL3, VSTM2L, [Histone family], IL15, PLCXD1, LOC101751545 [Cell-cycle CDK/cyclin], CRYBG3, CLEC3A, GUCA1B, TRNP1, UBE2T, UBE2K, CAMKK1, DTD2, ANKRD22, CLDN19
24.0-26.0 kDa	1045	2	18	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; Tubulin family; p53 pathway	high	, PGLS, ZFAND6, TCTEX1D4, EIF4H, PSMD10, FEV, NUDT16L1, CHMP2B, NKAIN3 [Actin family], MPZL2, TIMP4, AKAP14, IL2RA, HPRT1, CCDC59, GSTK1, LIN54
26.0-28.0 kDa	1054	3	21	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	RABL3, PPP1R1B, NKIRAS1 [Actin family], BATF, TSN, DHRS11, C1QB, BCL7B [Apoptosis caspase/BCL], DNAJB9, LOC121106941, TPPP [Tubulin family], NFU1, PPM1L, , , COQ10A, PPP1R2, COQ10B [Mitochondrial markers]
28.0-30.0 kDa	977	2	20	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	RCHY1, KIAA1644, LOC421419, UTP23, FOLR3, PTPRF, RTRAF, APOA1, RPL8, , TPRG1L, ZCCHC9, LDLRAD4, THYN1, , PEX11B, SHISA9, NQO1
30.0-32.0 kDa	1048	4	25	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Lamin family; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	TMEM241, , ASB13, ITM2B, TMEM220, DENND1A, MLEC, DRAM2, ATG5, HOXD12, TFAM [Mitochondrial markers], ELOVL3, SNAI2, RALB, ZFP511, AGPAT2, , LOC101747821
32.0-34.0 kDa	1101	5	33	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	MBP, SPAG16, THAP8, PHYHD1, PSMD14, KIAA1191, , FAM228, , MXD1, RALY, NECAP1, MTCH2 [Mitochondrial markers], HUS1, , GRAMD2, RSPO3, AQP3
34.0-36.0 kDa	1253	5	33	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	UNG, STUB1, MADPRT, SLC35E3, GSG1, RASSF3, ORC6, SYNDIG1, CD72L1, CCDC117, , ZNF277, RAX, CCNDBP1 [Cell-cycle CDK/cyclin], MTCH1 [Mitochondrial markers], SPDEF, SPARC, CD200R1A
36.0-38.0 kDa	1181	0	27	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	PNKD, CHIR3B3, LOC112533538, MTERF4 [Mitochondrial markers], SUSD4, AASDHPPT, TMEM189, USF1, CCN3 [Cell-cycle CDK/cyclin], IRF1, CD274, PPP1R3C, LUZP2, TOB1, LOC121112382, APTX, RP2, RCN1
38.0-40.0 kDa	1121	1	36	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	HSD11B1L, INSYN2B2, GALM, MAP2K6 [MAPK cascade], , NODAL, MAFK, SNX20, LDB2, SDF4, ALDOC, MAP2K6 [MAPK cascade], ELAVL1, RNF215, SLC35C2, LOC107050717, TCF7L2, B4GALT4
40.0-42.0 kDa	1184	2	39	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	MITF, PARP11, RBFOX2, PDGFRL, MEIS2, CPPED1, , SH3GL3, DMTF1 [Cell-cycle CDK/cyclin], , CFAP410, IL10RB, FAM45A, USP41, SH3GL2, ALS2, RPLP0, FGFR1OP
42.0-44.0 kDa	1092	2	42	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; p53 pathway	high	MEIS1, PRSS23, TMEM164, ATG4A, UBE3D, DEK, BGV, LOC100858953, ACTA2 [Actin family], ZDHHC2, GORAB [Actin family], ACTBL2 [Actin family], NR1I3, STK17B, B3GNT3, ACTC1 [Actin family], ERCC8, DPEP1
44.0-46.0 kDa	959	1	46	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	LAMP3, BUB3, B3GNT7, DCP2, PIPOX, MAPK9 [MAPK cascade], KLHL31, RPRD1B, SERPINE3, SAMD14, MYZAP, CDC37, SEH1L, NT5C3B, SERPINE2, OXTR, RHBDL3, TRAPPC13
46.0-48.0 kDa	1065	2	44	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	NTSR1, PDLIM7, PHYHIPL [Actin family], TMEM237, IDH2, UBXN6, LIMS1, LUC7L2, GLA, PAX6, SPOCK1, MAP2K5 [MAPK cascade], SS18, RSL1D1, SEC61A2, SCRN1, SMYD2 [Histone family], ABRAXAS1
48.0-50.0 kDa	1003	2	33	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family	high	SEPTIN6, U2SURP, KMT5A [Histone family], FADS1, WRAP73, RSRC2, STK24, GLUL, TTLL9 [Tubulin family], MTHFSD, KCTD18, SLC16A7, BZW1, DLST, DRD3, SYT13, RARB, PDE7B
50.0-52.0 kDa	976	2	28	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	TUBB6 [Tubulin family], SMARCD2, SMOC1, TRMU [Mitochondrial markers], FKBP10, BTBD10, RAD18, AZIN1, CYP2AB5, DLST [Mitochondrial markers], AS3MT, SHANK2, MAP9, STYK1, SLC29A3, CHRNA10, WIPI2 [Actin family], LOC121106919
52.0-54.0 kDa	935	0	22	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	POLL, PRR33, CRBN, RGS6, PPP2R2A, MRPS27, CPQ, CCDC15, KATNA1, LOC121108224, PTGER4, TCTN3, PPP2R2D, SCAI, GLRA2, DCTN4 [Actin family], RGS6, PARK2
54.0-56.0 kDa	957	0	44	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	TRMT13, SPNS3, PHGDH, SF3A3, IL17REL, HSPA13 [Heat shock proteins], SYT17, PLEKHD1, THUMPD2, HSPA14 [Heat shock proteins], SLC38A7, TUBE1 [Tubulin family], MGAT4F, MATN1, GTPBP3 [Mitochondrial markers], CCDC162, ENTPD8, MFSD5
56.0-58.0 kDa	915	2	16	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family	high	FTO, LOC107056139, ZBTB8A, KPNA3, TMEM39B, HCK, SH2D2A, HEPACAM2, CYP2AC1, TH, ALDH3B1, , LRRC71, ZNF606, DUS1L, TBLX1, CHRNB2, ENTPD8
58.0-60.0 kDa	877	0	27	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	GJA10, LACTB, PKM, KPNA2, LACTB, LIPI, SOAT1, PLXDC2, CAMK2B, DPYS, RFX5, CTTN [Actin family], HEXA, , SESN3, PFKFB1, CCDC77, TCF12
60.0-62.0 kDa	776	2	17	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	YRK, UGT1A1, ZFYVE28B, ATE1, , SRC, ASIC1, ESAM, C8B, LOC107050229, GPT2, LOC107050463, CAT, DPYS, FZD5, CABIN1, PLD5, PIGQ
62.0-64.0 kDa	681	1	14	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Mitochondrial markers; NF-kB/Rel; Tubulin family	elevated	SPATS2L, PLBD2, CARD9 [Apoptosis caspase/BCL], GN3L, CAMK2G, FAAH, MYEF2, PFKFB4, SLC43A2, GALNT14, ATL1, DKC1, NEFL, LMOD3, NEURL1, SNTA1, ACOD1L, BCO2 [Mitochondrial markers]
64.0-66.0 kDa	739	1	18	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; p53 pathway	high	KPNA5, ALB, KIZ, GPC5, ENTPD1, HABP2, IL2RB, GALNT18, INSC, TIMD4, UBQLN4, CNOT6, WFIKKN2, ENO4, ALAS1, CCDC88B, CCAR1 [Apoptosis caspase/BCL], LOC107056275
66.0-68.0 kDa	637	3	13	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel	high	RAP1GDS1, SACM1L, BFSP2, NTN4L, TDP1, KLHL10, RNGTT, SLC6A2, RPAP2, ENC1, IL1RAP, PTCHD3, KBTBD2, FBXW7, P2RX7, SLC2A1, FBXO21, SLC6A20
68.0-70.0 kDa	630	4	16	Actin family; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	SEPTIN9, RPA1, AIFM3, KLHL3, PTPN9, LOC419409, SLC18A2, GUCY1B3, , KLHL20, BFSP1, LOC101751434, ZNF606, LOC419429, SCAI, COLGALT1, SLC25A12, CHRDL2
70.0-72.0 kDa	575	2	11	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	MPP2, PRC1, RAB11FIP5 [Actin family], ARHGAP18, GGA1, TP73, ENTPD1, EYA4, ZBTB44, CROT, TAB3, ABLIM3 [Actin family], LOC420419, RUFY2, TRAF4, ZBTB7C, LOC101751279, ICK

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
72.0-74.0 kDa	537	3	12	Actin family; ER markers; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	GRAMD1C, TMEM62, RHOBTB1, RUFY1, NUMB, SLC6A14, HSPA5 [Heat shock proteins], MTMR6, RASGRP3, PRMT5, FRMD1, TP63, FOXP1, CAPN13, LDB3, CHFR, TESK2, RAD21
74.0-76.0 kDa	461	0	8	AKT pathway; Actin family; Heat shock proteins; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR	elevated	TRIM47, SPTY2D1, BRD7, DAGLB, BRD7, XRCC6, HEATR3, SCIN, SYK, IQUB, SCNN1G, C4BPS, SCML2, SIMC1 [Actin family], SLC25A13, TGFBR2, SLCO1A2, MTMR7
76.0-78.0 kDa	517	0	17	Actin family; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR	elevated	FIGNL2, ABCB7 [Mitochondrial markers], ADD1, TBC1D23, HBS1L, LOC420374, LIMK1, HNRNPM, RACGAP1, RHPN1, POLH, PCCA [Mitochondrial markers], FBLN1, TLE4, ZNF302, HOOK3, NEK8, MARCHF7
78.0-80.0 kDa	514	1	14	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR	elevated	FAM120B [Nuclear receptor PPAR], TNFAIP2, CD180, SNPH, LOC101750925, PARP12, CCDC110, LOC101751348, PATL2, FBXO34, DNAAF2, RTN1, GFM1 [Mitochondrial markers], HSPA8 [Heat shock proteins], CHST4, ALK, ARHGEF16, GTSE1
80.0-82.0 kDa	511	1	17	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	SLC7A4, SNAP91, MTA1, VIT, MARK1, LOC419333, , AKAP8L, MRE11, SSRP1, RTN1, MME, TRIM9, SEMA3B, LEKR1, SEC23B, RASEF, PFKP
82.0-84.0 kDa	501	4	14	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	CEP63, ANKRD13D, VIT, PCASP2 [Apoptosis caspase/BCL], VRTN, TRAK1, ARNT, SMYD4, USP49, IL23R, TLE4, FGFR4, CBARP, SLC26A6, RPS6KA2, BRD2, MARCHF7, MAPT
84.0-86.0 kDa	449	6	14	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers; p53 pathway	high	EVI5L, APBB2, MTSS1L, SGIP1 [Actin family], GTPBP4, TLE3, GPAT2 [Mitochondrial markers], IL4R, NCKIPSD [Actin family], JPH3, COG4, MARCHF7, HSP90AA1 [Heat shock proteins], ELMO1, KIF2A, LRFN5, LRRC1, RUBCNL
86.0-88.0 kDa	430	0	12	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; JAK-STAT; Mitochondrial markers; NF-kB/Rel; p53 pathway	elevated	DOCK2B, SIM1, MYBL1, MFN2, LOC107050560, MCMBP, C1R, PFKL, GRAMD1B, PPIG, EZH2 [Histone family], EFR3A, ARHGAP17, NELL2, ARMH3, PTPDC1, USP10, VPS51
88.0-90.0 kDa	399	0	13	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers	elevated	TTC27, CDH18, CLCN7, GCC1, UHRF1, KIFAP3, ATXN7, PDE10A, , VAV2, EPS15L1, LARGE1, PAMR1, RXFP1, THEMIS, LOXL3, ARHGAP10, PRDM4
90.0-92.0 kDa	360	0	13	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; p53 pathway	elevated	SLMAP, CSF2RA, ACAP2, PTPRA, PDZRN4, HECTD2, SLC13A3, C7, HPS6, POT1, ECE1, LRRN4, ADAM9L, EZH1 [Histone family], DTNA, , SMO, TRPC1
92.0-94.0 kDa	344	1	11	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; JAK-STAT; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family; p53 pathway	elevated	APBA1, SCYL2, FYB1, EPS8, SIDT1, DGKB, UHRF2, FGFR3, ATXN7, , GMIP [Actin family], , PROM1, INTS6, STRIP1 [Actin family], AFAP1 [Actin family], CFAP221, EML1
94.0-96.0 kDa	337	4	13	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; p53 pathway	high	ABR, AGBL2, HLCS, IKBKE [NF-kB/Rel], SLMAP, GPAM [Mitochondrial markers], LOC430766, SOBP, LRP3, STOX1, CCSER2, BACH2, ZC3H12B, ST14, MOBP, DRC7, PRLR [Actin family], ESYT3
96.0-98.0 kDa	359	2	17	Actin family; Apoptosis caspase/BCL; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	VAV3, RIN2, PLA2G4EL6, EXOC6B, DDR2L, ZNF438, BBX, , EPB41L1, AHR2, VWA2, SIK2, POSTN, PRKD1, EXOC6, HECTD3, BCAR3, FGD4
98.0-100.0 kDa	320	0	14	Actin family; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	elevated	FGD3, ENPP2, CTNNA3, LOC431300, SUN1, COL21A1, RASA3 [p53 pathway], ACO1, BEND3, TAOK3, ZNF598, TCOF1, CFAP69, ADAM19, EXOSC10, CDH1, SAFB2, CHRD
100.0-102.0 kDa	323	0	9	Actin family; Apoptosis caspase/BCL; JAK-STAT; MAPK cascade; Mitochondrial markers; p53 pathway	elevated	MYCBPAP, PDE6C, MTHFD1L [Mitochondrial markers], SEZ6, CARNS1, PSMD2, AARS2, SAFB2, GRM3, PPP6R2, ARVCF, IL31RA, WWP2, STAT2 [JAK-STAT], PTGFRN, CARNS1, FGD3, GRIA3
102.0-104.0 kDa	319	0	7	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel	elevated	UNC5C, CENPC, CTNNA2, LOC101749378, EIF4ENIF1, ARVCF, DPP9, NEDD4L, DLG1, TRPC6, ANAPC2, LGR4, NCOA7, KSR1, PPP1R12B, CHD1L, SYNE3, NRP2
104.0-106.0 kDa	282	0	9	Actin family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	elevated	SLC4A1, BCAN, ECT2, SLC26A8, INPP4B, FAM65C, N4BP1, NFKB2 [NF-kB/Rel], PPFIBP1, ZDBF2, GRM7, VPS54, FAM13B, RNF111, CHAF1A, PIGG, CLSTN2, ROR1
106.0-108.0 kDa	273	1	9	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; JAK-STAT; Lamin family; Mitochondrial markers; NF-kB/Rel	elevated	PHF14, NEDD4L, FAM65C, CWF19L2, MDGA1, PSMD1, FAM193B, TRIM24, DNAJC6, SH3D19, FRMD4B, EPB41, SPECC1, SYNE3, TMTC3, ANKLE2, KSR2, BCR
108.0-110.0 kDa	255	1	6	Actin family; Cell-cycle CDK/cyclin; Histone family; JAK-STAT; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	elevated	ST18, NEK10, NR3C2, PXN, EFTUD2, SEMA6A, EPB41L2, CORIN, CEP112, PCDH8, NCOA7, EML4, INO80D, NEDD4, COL19A1, AARS1, PSD3, MDGA1
110.0-112.0 kDa	240	4	9	Actin family; Cell-cycle CDK/cyclin; Histone family; JAK-STAT; Lamin family; Mitochondrial markers; Nuclear receptor PPAR	elevated	MMS19, NEDD4L, ATP1A1, ITGAV, PHF12, SEMA5A, INTS4, DGKZ, ANO8, LOC107050879, CDKL5 [Cell-cycle CDK/cyclin], SRBD1, PPP1R12A, ELAPOR2, MIB1, ITGA9, KIF5B, EPHA5
112.0-114.0 kDa	218	2	2	Actin family; Histone family; MAPK cascade; Mitochondrial markers; Tubulin family	elevated	IARS2 [Mitochondrial markers], EML4, LOC112530482, AARS1, LOC100859467, FHIP1B [Actin family], ZC3H4, ATP1A2, PTK2B, TRIM37, INPP4A, KCNH5, PCDH10, HIRA, KANSL1L, USP15, CRB1, CLIP2
114.0-116.0 kDa	235	1	4	Actin family; Cell-cycle CDK/cyclin; MAPK cascade; Mitochondrial markers; Tubulin family	elevated	UBAP2L, ZCCHC2, SEMA6A, EPB41L2, AP3B1, SAMD11, MKL2, PLEKHG5, ULK1, EML4, NHSL2, RUBCN, , HK3, ATP9A, GRID1, HPS3, PCDH10
116.0-118.0 kDa	208	2	2	Actin family; Apoptosis caspase/BCL; Histone family; MAPK cascade; Mitochondrial markers; Tubulin family	elevated	PLEKHG3, TDRD7, PLCL1, CARD10 [Apoptosis caspase/BCL], ACLY, TNK2, ITGA8, KLB, RECQL5, MEI1, KCNH8, MKL1, , ITGA8, TLR21, FBXO18, PSD3, OTUD4
118.0-120.0 kDa	228	0	5	Actin family; Apoptosis caspase/BCL; Histone family	elevated	KDM4C [Histone family], LIMCH1, XPO1, SLC12A5, ATP9A, FAM65A [Actin family], SLC12A1, FAM120C, LNPEP, DLGAP2, CASR, FRMD4A, OFCC1, TSHZ1, PLEKHM1, RPS6KC1, IDE, LIMCH1
120.0-122.0 kDa	207	0	3	Actin family; Histone family; Lamin family; Tubulin family	elevated	DIS3L, SEC24A, ADAMTS6, IPO7, TMPRSS9, HIP1R [Actin family], ANKRD52, USP8, PLEKHG5, PDGFRB, STARD8, SMARCA1, SSH1, B4GALNT3, LOC424167, SH3PXD2A, DIS3L2, GRID2IP [Actin family]
122.0-124.0 kDa	247	0	5	Actin family; Histone family; Lamin family; Nuclear receptor PPAR; Tubulin family	elevated	PIK3CB, PLCH2, MEGF10, PTK7, SUGP2, STARD8, PXN, SMARCC1, FNIP2 [Actin family], KIAA1522, SRGAP1, USP28, LRIG3, KDM4C [Histone family], LRRFIP1 [Actin family], MIS18BP1, STARD8, PHF2
124.0-126.0 kDa	179	1	3	Actin family; Histone family; JAK-STAT; Tubulin family; p53 pathway	elevated	XPO7, GLG1, ARHGAP39, ROBO2, ARHGAP39, SRGAP3, KIAA1522, ZNF451, UBE3C, GMIP [Actin family], MYO1H, BMP2K, ITGA2, ZRANB3, ARHGAP45, SLC38A10, ITGA7, ZRANB3
126.0-128.0 kDa	189	0	2	Actin family; Apoptosis caspase/BCL; Lamin family; p53 pathway	elevated	TNK2, ZNF628, MYO1C, CORIN, POLR3B, TTF2, CKAP2L, ADAR, LOC107050335, SIN3B, FARP1, ADNP, SLC4A7, B4GALNT4, TRPA1, CEP131, LAMB3 [Lamin family], TP53BP2 [p53 pathway]
128.0-130.0 kDa	192	0	4	Actin family; Apoptosis caspase/BCL; JAK-STAT; Lamin family; Mitochondrial markers; p53 pathway	elevated	, PIK3CG, DGKK, SLC4A5, SEC24B, SYNGAP1, FAM129A, ZFPM2, SENP6, SMC5, ATP2B4, PPIP5K2, ZMYND8, TOPORS, ATP13A5, MICAL1, COBLL1, AUTS2
130.0-132.0 kDa	180	0	10	Actin family; Apoptosis caspase/BCL; Lamin family; Tubulin family	elevated	MCF2L, RBM20, , FNIP1 [Actin family], PER3, CEP128, LOC121108097, IQSEC1, HEATR6, BLM, TMEM132A, ATP11A, DAB2IP [Actin family], PIK3CG, SORBS1, NRDE2, VWA3A, ZNF644
132.0-134.0 kDa	171	0	3	Actin family; Apoptosis caspase/BCL; JAK-STAT; Lamin family; NF-kB/Rel	elevated	GPR158, TAF2, PKP4, TIE1, RASGRF2, KCNT2, ADGRG6, GRM5, PLCB1, MYO1B, CWF19L1, ASTN1 [Actin family], COL1A1, MAN2A2, LOC112531052, CFAP61, STXBP5L, SASH1
134.0-136.0 kDa	174	0	9	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; JAK-STAT; Lamin family; Tubulin family	elevated	NCKAP1L, MCF2L, , CGN, LOC107050335, TAF2, WASHC5, ATP8B4, KCNT1, FBRSL1, ATP13A4, ZMYND8, CAPRIN2, ERBB2, SORCS3, CHL1, PRDM15, PHKA1
136.0-138.0 kDa	154	0	0	Actin family; Cell-cycle CDK/cyclin; Histone family; Lamin family; Tubulin family	elevated	RAPGEF1, , FBRSL1, KCNH7, WASHC4, ADAMTS3, TBC1D1, CNTLN, , PCDH7, PARD3, CFAP61, REXO1, SALL1, NFASC, MTMR4, ZNF687, TNKS
138.0-140.0 kDa	134	0	2	Actin family; Cell-cycle CDK/cyclin; Histone family; Lamin family; MAPK cascade; Tubulin family	elevated	EVC2, PCNX2, NOL6, DCTN1 [Actin family], DIAPH2, NRK, GRM5, PTPRC, MCF2L, RAPH1, JARID2, EGF, IARS, REV1, MAP1S, KIAA0825, LOC415478, SMC2
140.0-142.0 kDa	143	0	4	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; MAPK cascade; NF-kB/Rel	elevated	NEK1, KIAA0825, ABCB11, ATRNL1 [Actin family], BSN, MECOM [Histone family], KCNMA1, TJP2, SORBS1, BRIP1, KIF16B, TBC1D9, PPFIA4 [Actin family], PHLDB2, MAP3K6 [MAPK cascade], ADGRA2, MECOM [Histone family], ARHGEF10L
142.0-144.0 kDa	132	0	3	Actin family; Cell-cycle CDK/cyclin; MAPK cascade	elevated	CRAMP1, ATN1, NPC1, ERBB4, , TRPS1, DCTN1 [Actin family], MMS22L, ZMYM2, NRK, MYPN, HDLBP, IQSEC1, ARHGEF9, TBC1D32, USP42, MYBPC3, SPAG5
144.0-146.0 kDa	142	0	4	Actin family; Cell-cycle CDK/cyclin; Histone family; MAPK cascade; Tubulin family	elevated	SOS1, VEGFRKDRL, RRP12, MCF2L2, WDR33, DENND5B, DENND5B, ZNF521, FAM21C, STAG1, YEATS2, ZNF592, TPP2, PHLPP1, TBC1D4, LOC415478, ATP13A3, PLEKHA7
146.0-148.0 kDa	110	0	1	Actin family; Histone family; Tubulin family	elevated	MYO6, , IKBKAP, SCAF8, CLUH, ATRNL1 [Actin family], VEGFRKDRL, PPFIA2 [Actin family], CHD1W, IKBKAP, SSH2, PHLDB2, STK36, WDR17, SYNRG, TET1, EHMT1 [Histone family], CENPJ
148.0-150.0 kDa	131	0	6	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; MAPK cascade	elevated	PER2, NUP153, SCAF8, CNTNAP1 [Actin family], NEDD4, CADPS, MAPK8IP3 [Actin family; MAPK cascade], ZNF423, PLCG2, SOS1, RNF123, COL17A1, TTC21B, ZNF521, PHLPP2, KANK1, KIAA0825, RPGR
150.0-152.0 kDa	115	0	3	Actin family; Histone family; MAPK cascade	elevated	MAP3K1 [MAPK cascade], SIK3, ABCC4, ASXL2, NRXN1, SPATA13, KDR, MAP4K4, PDS5A, ANKFN1L, ATP8B2, SPATA13, ZNF521, BLM, ADCY9, IKBKAP, KTN1, LOC100858919
152.0-154.0 kDa	105	0	2	Actin family; Cell-cycle CDK/cyclin; Histone family; MAPK cascade	elevated	KIF24, MAP3K1 [MAPK cascade], PLEKHH1, ABCC4, NEO1, MEGF6, WDR7, UMODL1, MAP3K5 [MAPK cascade], NFASC, RGS22, CARMIL1, PNPLA7, TCP10, MAGI1, MIA2, TRAPPC8, PARD3
154.0-156.0 kDa	86	0	1	Actin family; Cell-cycle CDK/cyclin; Histone family; p53 pathway	routine	PLCG2, COL4A1, VPS8, TA3, NUP155, GRIN2A, PTPRJ, PTPRJ, TRIOBP [Actin family], LMO7, EDC4, SCAPER [Cell-cycle CDK/cyclin], CRB1, KDM6A [Histone family], COL4A3, MET, CEP164, TMEM131L
156.0-158.0 kDa	89	0	3	Actin family; Apoptosis caspase/BCL; Histone family	elevated	NCAPD3, IFT140, ADGRL2, PPIP5K1, ATRN [Actin family], PHLDB1, ATAD2, MEGF6, AQR, KTN1, FHOD3, CRB2, NPHP4, LOC424998, FANCD2, RIMBP2, MMR1L3, AKAP13
158.0-160.0 kDa	89	0	2	Actin family; Histone family; Mitochondrial markers	elevated	LMO7, PCMTD2, LTK, CLIP1, PTPRK, , TTC37, LTK, DISP2, SETD5, KDM6A [Histone family], CASKIN2, MED14, STARD13, MYOM3, LTK, USP19, CPS1 [Mitochondrial markers]
160.0-162.0 kDa	111	0	2	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family	elevated	KDM6A [Histone family], ESPL1, UACA, MED14, DISP1, SYNGAP1, CRB2, PTPRU, GLI3, A2ML2, KIF7, BCL9L [Apoptosis caspase/BCL], RAPGEF2, ZNF831, COL4A5, , ZNF423, SMG6
162.0-164.0 kDa	94	0	3	Actin family; Cell-cycle CDK/cyclin; p53 pathway	elevated	SBNO2, DISP2, LMTK2, TULP4, PTPRK, CRACD [Actin family], ARHGAP5, PTPRM, A2ML2, ZNF608, EEA1, WDR7, ABCC6, CEP162, SLIT3, ABCC8, CECR2, PTPRM
164.0-166.0 kDa	96	0	3	Actin family; Apoptosis caspase/BCL; MAPK cascade; Mitochondrial markers	elevated	A2M, SETD5, KIF24, MMR1L4, ABI3BP, SOGA1, AQR, CRACD [Actin family], CPS1 [Mitochondrial markers], MYOM2, BCL9L [Apoptosis caspase/BCL], MYOM2, NRXN1, CAMSAP2, IQGAP2, PTPRU, NKTR, NKTR

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
166.0-168.0 kDa	100	0	6	Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; p53 pathway	elevated	CLASP1, DOT1L [Histone family], TRAPPC8, RIMS2, POLA1, COL4A2, ABI3BP, RIMS2, ADGRL2, SCRIN1, HECW2, ITSN1, OVST, EEA1, , CLASP2, EPRS, ABI3BP
168.0-170.0 kDa	72	0	1	Actin family; Histone family	routine	CLASP2, DOT1L [Histone family], POLA1, RALGAPB, TMEM131L, ABCC2, ADGRL3, RIMS2, BCORL1, CLASP2, PTPRM, CAMTA1, CLASP1, PTPRM, CFTR, KDM5B [Histone family], SLIT3, CAMSAP2
170.0-172.0 kDa	64	0	2	AKT pathway; Actin family	routine	PIK3C2G, ARHGEF12, RIMS2, MAGI2, COL24A1, BAZ1B [AKT pathway], ANKRD31, THSD7B, TOP2B, TULP4, MUC5ACL, ABCC1, GEMIN5, CRACD [Actin family], FHOD3, FGD5, PTPRS, DIP2B [Actin family]
172.0-174.0 kDa	53	0	0	Actin family; Histone family	routine	SLIT3, CCDC180, SLIT3, SYNJ1, ARHGAP5, MAGI2, LTBP1, COL1A4, LRP6, DIP2A [Actin family], TMEM131L, COL24A1, NHSL1, ZNF518A, ARHGEF12, ABCC1, COL11A1, TOP2A
174.0-176.0 kDa	59	0	0	Actin family; Lamin family	routine	ADGB, TARBP1, NRXN3, SYNJ1, COL1A4, ADAMTS12, TNS3, TULP4, COL1A4, AQR, COL1A4, , ABCA9, AGL, AGL, UGGT1, BRCA1, ABCC9
176.0-178.0 kDa	56	0	1	Actin family; Histone family; Lamin family; MAPK cascade	routine	EIF4G3, HECW2, CCDC141, , KDM5B [Histone family], MAP3K4 [MAPK cascade], COL5A1, ROBO1, RUSC2, LAMC1 [Lamin family], BAZ1A, COL11A1, COL16A1, DIP2A [Actin family], RIMS1, SMARCA2 [Actin family], C4, CEP170B
178.0-180.0 kDa	67	0	1	Actin family; Lamin family; MAPK cascade	routine	CTTNBP2 [Actin family], NHSL1, SMARCA2 [Actin family], HEG1, NES, , COL11A1, MAP3K4 [MAPK cascade], FAM208A, RREB1, CEP170B, SORL1, TIAM1, KIF14, RREB1, PAPPA2, FHOD3, PBRM1
180.0-182.0 kDa	63	0	1	MAPK cascade	routine	ABCA8, CEP170B, PLCH2, EIF4G3, LRP6, VTG3, ABCC9, EIF4G1, PXDN, URB2, WDR62, IQGAP3, ROBO1, ABCA9, , DLEC1, ROBO1, LTBP1
182.0-184.0 kDa	50	0	2		routine	COL11A1, ALS2, HEG1, SYNM, EIF4G1, SHANK3, CLTCL1, SAMD9L, ADGB, EIF2AK4, KIF21A, THOC2, ERCC6L2, HECW1, SCRIB, EIF4G3, RB1CC1, CAMTA1
184.0-186.0 kDa	58	0	0	Lamin family; MAPK cascade	routine	THOC2, ZCCHC11, PLCH1, SHANK3, LAMB2 [Lamin family], MYO15L, VTG3, KIF14, ARID2, RAPGEF6, SHANK3, USP6, CAMTA1, KIF21A, CAMTA1, THSD7A, CCDC162, POLR1A
186.0-188.0 kDa	37	0	0		routine	, MON2, MBD5, , , LTBP2, NES, ABCA5, NRCAM, PCDHGC5, BCOR, COL11A1, PCF11, , PLCH1, THSD7A, THSD7A,
188.0-190.0 kDa	49	0	0	Actin family; Histone family; Lamin family	routine	CFAP74, LOC101747860, PRDM2, EPRS, CACNA1I, FRMPD3, ARID2, MARF1, KIDINS220 [Actin family], PHRF1, CFAP43, AIM1, CLTC, BCOR, ZC3H13, VTG2, SIPA1L2, ARAP2
190.0-192.0 kDa	33	0	2	Actin family; Lamin family	routine	SHPRH, CDC42BPA, ITSN1, CLTCL1, FOCAD, COL11A1, TMEM131L, FAM208A, PLCH1, IQGAP2, PCDH15, SIPA1L2, ADAMTSL1, SLX4, ARFGEF2, NRAP, EIF4G3, NRAP
192.0-194.0 kDa	50	0	0	Actin family; Lamin family; p53 pathway	routine	COL14A1, USP54, ARHGEF28, FMN2, CDC42BPB, RICTOR, TIAM2, SCRIB, CLTC, KDM5A, BCOR, CCDC162, KIDINS220 [Actin family], MYO3A, FAM208A, DOCK2B, LOC101748153 [Lamin family], ANK1
194.0-196.0 kDa	63	0	0	Actin family; Lamin family; p53 pathway	routine	CHD5, CDC42BPB, ADAMTSL3, PATJ, NUP98, RICTOR, SHROOM2, , ARHGEF5, , CFAP43, LRP4, ARAP3, NALCN, KDM3B, FRMPD4, QSER1, NUP188
196.0-198.0 kDa	48	0	0	Actin family; Lamin family; Tubulin family	routine	CEP152, TRPM3, LOC101747860, DOCK2B, SHROOM2, GCC2, ADAMTSL3, VIRMA, DOCK2B, NRAP, KIDINS220 [Actin family], CHD1W, PLXNA2, UNC13A, LAMB1 [Lamin family], PEAK1, AIM1, TNRC6A
198.0-200.0 kDa	46	0	0	Actin family; Lamin family; Tubulin family; p53 pathway	routine	TRPM3, TNRC6C, NALCN, CHD5, LOC101749352, NUP188, LOC101748153 [Lamin family], LAMA4 [Lamin family], LOC421441, ADAMTSL3, NUP210L [p53 pathway], PATJ, THADA, KALRN, ITSN2, ADAMTSL3, KIDINS220 [Actin family], ADAMTSL3
200.0-202.0 kDa	54	0	0	Actin family; p53 pathway	routine	UBR2, TJP1, CACNA1I, EPB41L1, NALCN, ADAMTS7, ADAMTSL1, PCSK5, PPL, UNC13A, CHD1W, FOCAD, ARID2, NRCAM, TANC1, KIF1A, VWDE, UBR1
202.0-204.0 kDa	47	0	1	Lamin family	routine	CDC42BPA, KIAA1549, TSC2, ARHGEF5, TANC1, LAMB4 [Lamin family], MAST2, LOC416924, CHD5, KIAA1671, LAMA4 [Lamin family], KIAA1549, TMEM131, COL14A1, TMEM131, EXPH5, IFT172, KNDC1
204.0-206.0 kDa	46	0	1	Actin family; p53 pathway	routine	MAST2, PLXNA2, MAST2, SHANK2, KIF26A, PRG4, GSE1, ECPAS, PPL, UBR3, ZNF236, EXPH5, TSC2, KIF1B, KNDC1, LARP4B, VTG2, TANC1
206.0-208.0 kDa	37	0	0	Actin family	routine	BTAF1, , VTG2, PHIP [Actin family], LOC101747860, LOC424300, CPAMD8, MTCL1, KIF26B, PSME4, PATJ, RAI1, PATJ, CACNA1S, CDC42BPA, ARID1A, ZNF106, AK9
208.0-210.0 kDa	55	0	0	Histone family; p53 pathway	routine	ARHGAP21 [p53 pathway], NUP214 [p53 pathway], PCSK5, TAF1, MAST2, TCF20, SCRIB, GBF1, ARFGEF1, ITGB4, CHD1, UBR3, GBF1, KAT6A [Histone family], TCF20, TNRC6A, ANK3, PLXND1
210.0-212.0 kDa	65	0	0	Histone family; p53 pathway	routine	KIF26A, MTCL1, SCN9A, SCN8A, MAST2, KIF26A, PLXNA2, VWA8, FER1L6, UBR3, SVIL, LOC421255, ZNF236, LMO7, Mylk, RESF1, SBF2, ANK3
212.0-214.0 kDa	47	0	0	Actin family; p53 pathway	routine	KIAA1217, PSME4, LRP4, ANK2, ZNF236, NUP214 [p53 pathway], AGRN, SHROOM3, THADA, NAV1, MYO5A, PLXNA4, MAP2, WDR81, CCDC88A, KIF26B, MYO5A, KIF26A
214.0-216.0 kDa	57	0	0	p53 pathway	routine	SBF2, TNC, SCN8A, PTPRD, MYH9, MAST2, ADAMTS20, ANK3, MPDZ, SHROOM3, DOCK2, MYO16, OTOF, WDR81, PLXND1, ANK1, DOCK1, SDK1
216.0-218.0 kDa	29	0	1	Actin family	routine	ARID1A, RLF, GREB1L, CCDC88A, NCOA6, Xirp1 [Actin family], THADA, ANAPC1, RALGAPA2, KIF16B, KIF13A, SCN1A, SHROOM3, AKAP12, KIF20B, DENND4C, AKAP12, KNDC1
218.0-220.0 kDa	41	0	1	Cell-cycle CDK/cyclin	routine	TANC2, MYLK, DOCK4, LCT, CHD4, MYO5A, SBF1, CHD4, CCDC88A, NAV1, MALRD1, KIF26A, SBF1, LOC421441, TNC, DICER1, DICER1, SBF1
220.0-222.0 kDa	37	0	6	Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; p53 pathway	elevated	RBBP6, ANK3, CASP8AP2 [Apoptosis caspase/BCL], ANKRD12, GREB1L, AGRN, HELZ, CASP8AP2 [Apoptosis caspase/BCL], CDK5RAP2 [Cell-cycle CDK/cyclin], NCKAP5, PHF3, MYO7L2, MYH9, TET2, LOC395944, CDK5RAP2 [Cell-cycle CDK/cyclin], RALGAPA2, PRR14L
222.0-224.0 kDa	44	0	2	Cell-cycle CDK/cyclin; Histone family; p53 pathway	routine	DSCAM, ADAMTS9, MYH9, DSCAM, , ARHGAP21 [p53 pathway], TANC2, MYH15, ARHGAP21 [p53 pathway], NIN, ARHGAP21 [p53 pathway], DOCK5, PCM1, KIF13A, ARHGAP33, MYH1F, MYH1F, MYH1B
224.0-226.0 kDa	38	0	2	Apoptosis caspase/BCL; Cell-cycle CDK/cyclin	routine	FANCM, EML5, MYH15, DSCAM, MYLK, MYH7B, AGRN, LOC395944, DOCK4, HEATR5A, HEATR5B, MYH9, TRIP11, MYH7, , MYH7B, CDK5RAP2 [Cell-cycle CDK/cyclin], SCN2A
226.0-228.0 kDa	34	0	1	Cell-cycle CDK/cyclin; Histone family	routine	KAT6B [Histone family], MALRD1, DOCK7, TTC3, TANC2, LOC395944, PLCE1, CKAP5, MYO7L2, PLCE1, ASXL3, MYH9, GTF3C1, MYH9, TSPOAP1, PCNX2, KIF26B, CDK5RAP2 [Cell-cycle CDK/cyclin]
228.0-230.0 kDa	33	0	0	Actin family	routine	MYH11, SCN3A, SCN3A, SCN2A, , NUP205, ARHGAP33, SCN1A, ASXL3, TNC, NCKAP5, TRIP12, NUMA1, SDK1, MYH11, TANC2, MYO5A, MYH7
230.0-232.0 kDa	23	0	0		routine	PHF3, EVPL, MYH10, TSPOAP1, DOCK4, SNRNP200, PCNX2, MED12, DLG5, SCN5A, LOC101751678, SORL1, EVPL, IGFN1, DOCK4, DOCK8, SETD1A, MYH10
232.0-234.0 kDa	30	0	0	Actin family	routine	SCN5A, AGRN, LOXHD1, PHF3, DOCK11, CCDC88A, AGRN, MYH10, MYH10, LOC420486, MYO9B, AGRN, TSPOAP1, CIT [Actin family], ATG2A, PDE4DIP [Actin family], ABCA1, BAZ2B
234.0-236.0 kDa	25	0	0	Actin family	routine	BAZ2B, KIAA0100, MYOF, FREM1, DOCK3, ANK2, CLIP1, ARFGEF3, SPEF2, NAV3, SDK1, MYOF, LOXHD1, CCDC88A, NUMA1, MYO9B, LOC101751678, LOC421441
236.0-238.0 kDa	34	0	0		routine	MUC5ACL, DOCK9, SYTL2, BAZ2B, MYO10L, DOCK9, MYOF, PCNX2, CACNA1C, CR1, PTPRB, MYOF, DOCK11, SDK2, ANKRD26, NAV3, TECTA, MYO9B
238.0-240.0 kDa	32	0	2	Cell-cycle CDK/cyclin	routine	MALRD1, CCDC88A, DYSF, MED12L, , BAZ2B, ASXL3, SPAG17, CDK5RAP2 [Cell-cycle CDK/cyclin], PI4KA, IGFN1, DOCK8, CDK5RAP2 [Cell-cycle CDK/cyclin], CABIN1, SDK2, DOCK7, NIN, SDK2
240.0-242.0 kDa	34	0	0	Actin family; Lamin family	routine	PIKFYVE, DOCK9, ABCA7, NUMA1, MED13, SDK1, DYSF, LOC101751678, MICAL3 [Actin family], DYSF, CACNA1D, KIAA1549L, WNK2, WNK2, SDK2, USF3, MED12, PIKFYVE
242.0-244.0 kDa	24	0	0	Actin family	routine	MED12, ARFGEF3, ASXL3, DOCK7, CLIP1, CACNA1C, DYSF, RALGAPA1, CACNA1D, DYSF, MED12, ARID1A, FRMPD2, PDE4DIP [Actin family], GTF3C1, , PRRC2B, C2CD3
244.0-246.0 kDa	21	0	0	Actin family	routine	CACNA1I, WNK2, ARHGEF4, SDK1, PLXNB3, MICAL3 [Actin family], PDE4DIP [Actin family], PLCE1, CACNA1D, ANKRD17, SVIL, ANKRD17, CABIN1, ARID1B, POLE, GON4L, ARFGEF3, INTS1
246.0-248.0 kDa	19	0	0		routine	SORL1, FREM1, ARFGEF3, CACNA1I, NAV3, WNK1, ABCA1, CLIP1, ABCA4, PRRC2B, OBSL1, FN1, PRR14L, IGFN1, CACNA1G, EVPL, PTPRB, PLXNB3
248.0-250.0 kDa	20	0	0		routine	CACNA1D, SPAG17, CACNA1G, TET1, WNK2, PLXNB3, YLPM1, CACNA1D, PLCE1, IGFN1, ANKRD26, TET1, SDK1, CACNA1D, KNTC1, AKAP6, RTTN, ARHGEF4
250.0-252.0 kDa	20	0	0	Actin family; Heat shock proteins	routine	MICAL3 [Actin family], MICAL3 [Actin family], NAV3, MICAL3 [Actin family], AKAP6, NAV3, MICAL3 [Actin family], NAV2, SDK1, POLE, RIF1, PLCE1, PKDREJ, LOC415414, NAV3, MYO7A, DNAJC13 [Heat shock proteins], ascc3
252.0-254.0 kDa	17	0	0	Actin family; Heat shock proteins	routine	CCDC162, CNTRL, NUMA1, C2CD3, ARHGEF4, NAV3, LOC415414, MICAL3 [Actin family], LOC415414, HEATR1, ARHGEF4, MYO7A, LOC101751678, ZNF407, CACNA1E, AHCTF1, DNAJC13 [Heat shock proteins]
254.0-256.0 kDa	14	0	0	Actin family; Heat shock proteins	routine	ABCA4, ABCA1, MICAL3 [Actin family], KIAA0100, RIF1, RIF1, ABCA1, DNAJC13 [Heat shock proteins], KNTC1, DNAJC13 [Heat shock proteins], IGFN1, ZNF462, PTPRZ1, SPTB
256.0-258.0 kDa	15	0	0	Heat shock proteins	routine	SVIL, DNAJC13 [Heat shock proteins], ARHGEF4, EIF4EBP3, C2CD3, CAD, NOTCH2, NOTCH2, CAD, NOTCH2, SEC16A, PTPRZ1, CENPE, DOP1B, ARHGEF4
258.0-260.0 kDa	22	0	0		routine	CSPG4, SVIL, NAV2, CADNL, NAV2, NOTCH2, PKDREJ, ABCA2, TTC28, URB1, CACNA1E, PTPRZ1, PTPRVP, TTC28, MYO7A, SPTB, SVIL, GOLGA4
260.0-262.0 kDa	19	0	0	Histone family	routine	JMJD1C, APC2, EIF4EBP3, CELSR3, NAV2, EP300 [Histone family], ARHGEF4, POLE, AKAP6, ROS1, ROS1, PCNX1, SPTBN2, CACNA1G, BRWD1, FASN, IGFN1, LOC416696
262.0-264.0 kDa	25	0	0	Histone family	routine	GOLGA4, ACACA, PTPRVP, CACNA1B, NAV2, LOXHD1, NOTCH2, IGF2R, NOTCH2, ANKRD17, ARHGEF4, SPTB, NBAS, CADNL, ZNF462, ACAC, ANKRD17, CREBBP [Histone family]
264.0-266.0 kDa	16	0	0	Actin family; Histone family	routine	CREBBP [Histone family], NAV2, MAST4, FAM208B, NAV2, SEC16A, AKAP6, MPRIP [Actin family], MAST4, TLN1, ACACA, GOLGA4, DIDO1, SPTBN1, CACNA1B, MAP1B
266.0-268.0 kDa	23	0	0	Histone family; Lamin family	routine	NCOR2, MAST4, TTC28, GOLGA4, SEC16A, MAST4, MKI67, ARHGEF4, EP300 [Histone family], CSPG4B [Lamin family], NBAS, NBAS, CNOT1, DOPEY2, ACACA, NBAS, STAB1, TLN2
268.0-270.0 kDa	16	0	0	Histone family; Lamin family	routine	URB1, EP300 [Histone family], HIVEP3, IGSF10, MAST4, , TNRC18, MAST4, FLNB [Lamin family], CREBBP [Histone family], JMJD1C, NBAS, CNTRL, PDZD2, ATRX, EIF4EBP3

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
270.0-272.0 kDa	15	0	0	Lamin family	routine	TTC28, EIF4EBP3, CSPG4B [Lamin family], PCNX1, JMJD1C, ATRX, LRRK2, CACNA1E, JMJD1C, PDZD2, CACNA1H, TLN2, ACACA, TLN1, TLN1
272.0-274.0 kDa	16	0	0	Lamin family	routine	PKD1L2, LRRK2, SPTBN1, ANKRD17, PDZD2, CACNA1E, FLNB [Lamin family], ATRX, CHD9, FN1, ABCA2, FAM208B, DOPEY2, PRPF8, SPG11, TLN1
274.0-276.0 kDa	15	0	0		routine	SPTBN1, NCOR2, FAM208B, STAB2, NCOR2, ATRX, ABCA2, FASN, , DIDO1, ATRX, NOTCH1, PRPF8, CACNA1H, IGF2R
276.0-278.0 kDa	19	0	0		routine	NCOR1, MAP1B, FAM208B, NOTCH1, FAM208B, MAST4, TPR, FAM208B, FAM208B, ZNF462, CEP192, PRPF8, STAB2, HECTD1, SPTAN1, JMJD1C, CADNL, HECTD1
278.0-280.0 kDa	18	0	0		routine	DOPEY2, EIF4EBP3, FAM208B, PTPN13, JMJD1C, CEP290, FASN, STAB1, GVINP1, SON, JMJD1C, JMJD1C, EIF4EBP3, DOPEY2, HIVEP2, JMJD1C, MAST4, SPTAN1
280.0-282.0 kDa	20	0	0	Lamin family	routine	ANKRD26, POLQ, IGF2R, MKI67, DOPEY2, FLNB [Lamin family], CADNL, ZNF462, ABCA2, ABCA2, FLNB [Lamin family], PIEZO1, HELZ2, FASN, IGF2R, FBN3, SPTAN1, LOC101751678
282.0-284.0 kDa	10	0	0	Lamin family	routine	MAP1B, MKI67, USP24, TENM3, MAST4, NCOR1, , , STAB1, FLNB [Lamin family]
284.0-286.0 kDa	16	0	0	Actin family; Histone family	routine	ANKRD26, ZNF462, EIF4EBP3, PIEZO1, SETD2 [Histone family], ALMS1, XIRP1 [Actin family], , LRRK2, SPTAN1, USP9, AKAP13, HECTD1, ZNF292, XIRP1 [Actin family], ANKRD26
286.0-288.0 kDa	13	0	1	Histone family	routine	HECTD1, CRYBG3, ITPR2, SPTAN1, MTOR, WNK1, SPTAN1, SPTAN1, LOC101751678, FBN1, SETD2 [Histone family], , AKAP13
288.0-290.0 kDa	15	0	0		routine	EPG5, RALGAPA1, USP24, TG, CEP290, TENM4, PRRC2C, CELSR3, LOC427025, TENM2, VWF, TENM3, AKAP13, PRRC2C, AKAP13
290.0-292.0 kDa	15	0	0	Histone family	routine	USP9, , ZFYVE26, TENM3, TENM2, LOC107050955, CELSR2, RALGAPA1, HIVEP1, NIPBL, LOC101751678, UNC79, SETD2 [Histone family], LOC101751678, HIVEP1
292.0-294.0 kDa	13	0	0		routine	EPG5, FBN1, USP24, , TENM1, TENM2, PRRC2C, HIVEP1, PRRC2C, WNK1, GCN1, USP24, ANKRD11
294.0-296.0 kDa	7	0	0		routine	HIVEP1, ATR, MUC5ACL, HIVEP1, MYO9A, COL7A1,
296.0-298.0 kDa	11	0	0		routine	FBN2, FBN2, LOC121110182, RALGAPA1, LOC417131, RALGAPA1, MYO9A, APC, ITPR3, HIVEP1, MYO9A
298.0-300.0 kDa	9	0	0		routine	TENM3, FBN1, CELSR3, LOC121110182, CELSR3, LOC121110182, , OTOGL, FBN1
300.0-302.0 kDa	9	0	0		routine	CEP250, MUC5ACL, MYO9A, FBN2, LOC101751678, ITPR2, LOC427025, CEP250, ATR
302.0-304.0 kDa	13	0	0		routine	TENM1, TENM1, CHD6, TENM1, TENM2, MYO9A, WNK1, TENM3, ITPR3, TENM1, NIPBL, UBR5, UNC79
304.0-306.0 kDa	14	0	0		routine	ITPR3, CEP250, ANKRD11, ITPR3, UNC79, , NBEAL2, TG, NBEAL1, NBEAL1, ITPR2, NBEAL2, NBEAL2, VWF
306.0-308.0 kDa	10	0	0		routine	CHD6, NBEAL1, TENM2, NBEAL2, VWF, ITPR2, TENM2, TG, APC, APC
308.0-310.0 kDa	12	0	0		routine	TENM3, ANKRD11, UBR5, PIEZO2, TENM4, NBEAL2, TENM2, TENM3, , SETX, MXRA5, TENM4
310.0-312.0 kDa	15	0	0	Histone family	routine	UBR5, ITPR1, NSD1 [Histone family], TENM3, SETX, MYO18B, LOC414835, ITPR1, TENM2, TENM2, ITPR1, MUC5ACL, ITPR1, TENM3, FBN1
312.0-314.0 kDa	6	0	0		routine	TNRC18, VWF, APC, APC, TENM4, SETX
314.0-316.0 kDa	11	0	0		routine	TACC2, PIEZO2, RP1, DMXL2, TACC2, NIPBL, ITPR1, BPTF, MXRA5, MGA, PIEZO2
316.0-318.0 kDa	8	0	0		routine	TACC2, ITPR1, , TNRC18, , ALMS1, RP1,
318.0-320.0 kDa	11	0	0		routine	, TACC2, BPTF, PIEZO2, NF1, NF1, HTT, PIEZO2, UTP20, TRANK1, MAP1A
320.0-322.0 kDa	8	0	0		routine	MGA, LRBA, PIEZO2, BOD1L1, PIEZO2, NF1, HELZ2, BOD1L1
322.0-324.0 kDa	5	0	0		routine	PIEZO2, LOC429249, LRBA, MXRA5, NBEA
324.0-326.0 kDa	3	0	0		routine	CHD9, TACC2, DNAH5L
326.0-328.0 kDa	5	0	0		routine	COL6A3, DSP, COL6A3, NBEA, NBEA
328.0-330.0 kDa	6	0	0		routine	BOD1L1, NBEA, NBEA, BPTF, BOD1L1, BPTF
330.0-332.0 kDa	3	0	0		routine	DSP, , CELSR3
332.0-334.0 kDa	3	0	0		routine	COL12A1, COL12A1, ATM
334.0-336.0 kDa	13	0	0	Histone family; Lamin family	routine	FRY, ASH1L [Histone family], BPTF, KALRN, CELSR3, ZZEF1, COL12A1, LAMA1 [Lamin family], LAMA1 [Lamin family], COL6A3, EP400, KALRN, CELSR3
336.0-338.0 kDa	13	0	0	Histone family	routine	LOC776067, LOC776067, FRY, LOC776067, LOC776067, LOC776067, FRY, CHD7, EP400, CELSR3, ASH1L [Histone family], EP400, KALRN
338.0-340.0 kDa	18	0	0	Lamin family	routine	CHD7, FRY, LOC776067, KALRN, LOC776067, LAMA3 [Lamin family], FRY, LOC776067, LOC776067, LOC776067, LAMA3 [Lamin family], COL6A3, COL6A3, FRYL, DMXL2, LAMA1 [Lamin family], TRANK1, VPS13A
340.0-342.0 kDa	10	0	0	Lamin family	routine	DMXL1, COL12A1, LAMA1 [Lamin family], COL12A1, FRYL, HTT, BPTF, C1HXORF59, CDH23, CENPF
342.0-344.0 kDa	6	0	0	Lamin family	routine	DMXL2, DMXL1, LAMA3 [Lamin family], LAMA3 [Lamin family], HTT, FRY
344.0-346.0 kDa	5	0	0	Lamin family	routine	TRIO, FRYL, TRANK1, VPS13A, LAMA2 [Lamin family]
346.0-348.0 kDa	6	0	0		routine	TRIO, OTOG, , TRIO, FRY, MAP1A
348.0-350.0 kDa	9	0	0	Lamin family	routine	FRYL, COL6A3, LAMA2 [Lamin family], MAP1A, MGA, C1HXORF59, CEP350, LAMA3 [Lamin family], ATM
350.0-352.0 kDa	9	0	0	Lamin family	routine	, VPS13A, TRIO, DCHS1, LAMA3 [Lamin family], CELSR3, FREM2, CPLANE1, DCHS1
354.0-356.0 kDa	4	0	0		routine	C1HXORF59, SPEG, CEP350, MAP1A
356.0-358.0 kDa	8	0	0	Lamin family	routine	REV3L, MAP1A, , CFAP54, WDFY4, VPS13A, LAMA2 [Lamin family], CMYA5
358.0-360.0 kDa	3	0	0	Actin family; Lamin family	routine	CFAP54, LAMA2 [Lamin family], MACF1 [Actin family]
360.0-362.0 kDa	4	0	1	NF-kB/Rel	routine	VPS13A, RELN [NF-kB/Rel], CDH23, VPS13A
362.0-364.0 kDa	3	0	0		routine	VPS13A, SPEG, GOLGB1
364.0-366.0 kDa	2	0	0		routine	CDH23, UNC80
366.0-368.0 kDa	2	0	0		routine	, SVEP1
368.0-370.0 kDa	4	0	0	Lamin family	routine	SVEP1, LAMA2 [Lamin family], LAMA2 [Lamin family], CDH23
370.0-372.0 kDa	3	0	0		routine	UNC80, LOC121109863, UNC80
372.0-374.0 kDa	3	0	0		routine	UNC80, VCAN, SPEG
374.0-376.0 kDa	3	0	0		routine	NEB, , LOC101751678
376.0-378.0 kDa	4	0	1	NF-kB/Rel	routine	RELN [NF-kB/Rel], PTPRQ, BRCA2, PTPRQ
378.0-380.0 kDa	3	0	0		routine	ABCA12, BSN, PTPRQ
380.0-382.0 kDa	2	0	0		routine	, OTOG
382.0-384.0 kDa	2	0	1	NF-kB/Rel	routine	NEB, RELN [NF-kB/Rel]
386.0-388.0 kDa	4	0	1	NF-kB/Rel	routine	DNAH8, CSMD1, RELN [NF-kB/Rel], CSMD3
388.0-390.0 kDa	4	0	2	NF-kB/Rel	routine	VCAN, RELN [NF-kB/Rel], RELN [NF-kB/Rel], CSMD1
390.0-392.0 kDa	3	0	0		routine	SMG1, ZFHX4, CSMD3
392.0-394.0 kDa	3	0	0		routine	CSMD2, WDFY3, WDFY3
394.0-396.0 kDa	9	0	0	Heat shock proteins	routine	VPS13C, ZFHX4, HSPG2 [Heat shock proteins], HSPG2 [Heat shock proteins], HSPG2 [Heat shock proteins], HSPG2 [Heat shock proteins], ASPM, SZT2, WDFY3
396.0-398.0 kDa	4	0	0		routine	, UTRN, ZFHX4, UTRN
398.0-400.0 kDa	4	0	0		routine	CUBN, ASPM, UTRN, ZFHX3
400.0-402.0 kDa	5	0	0	Heat shock proteins	routine	ZFHX3, HSPG2 [Heat shock proteins], CUBN, SVEP1, SMG1
402.0-404.0 kDa	5	0	0	Histone family	routine	VPS13C, CSMD3, PCNT, KMT2A [Histone family],
404.0-406.0 kDa	6	0	0	Lamin family	routine	PCNT, USP34, LAMA5 [Lamin family], CSMD3, LAMA5 [Lamin family], DMD
406.0-408.0 kDa	5	0	0		routine	PCNT, PCNT, BSN, SZT2, VPS13C
408.0-410.0 kDa	1	0	0		routine	DMD
410.0-412.0 kDa	5	0	0		routine	USP34, SMG1, , LYST, NEB
412.0-414.0 kDa	5	0	0	Histone family	routine	SZT2, KMT2A [Histone family], KMT2A [Histone family], DMD, ANK2
416.0-418.0 kDa	1	0	0		routine	VPS13C
418.0-420.0 kDa	2	0	0		routine	ANK2, VPS13C
420.0-422.0 kDa	5	0	0		routine	LYST, DMD, DMD, ANK2, ANK2
422.0-424.0 kDa	7	0	0	Heat shock proteins	routine	VPS13C, HSPG2 [Heat shock proteins], DMD, DMD, DNAH12, ANK2, DMD

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
424.0-426.0 kDa	2	0	0		routine	DNAH7, DMD
426.0-428.0 kDa	3	0	0		routine	DMD, LYST, DMD
432.0-434.0 kDa	2	0	0	Heat shock proteins	routine	HSPG2 [Heat shock proteins], FRAS1
434.0-436.0 kDa	4	0	0		routine	FRAS1, FRAS1, TRRAP, PKHD1
436.0-438.0 kDa	2	0	0		routine	TRRAP, TRRAP
438.0-440.0 kDa	3	0	0		routine	VPS13B, ZNF469, FAT2
440.0-442.0 kDa	3	0	0		routine	FRAS1, PKHD1, FRAS1
442.0-444.0 kDa	3	0	0	Heat shock proteins	routine	HSPG2 [Heat shock proteins], DNAH8, PCNT
444.0-446.0 kDa	1	0	0		routine	NEB
446.0-448.0 kDa	1	0	0		routine	VPS13B
448.0-450.0 kDa	4	0	0		routine	DNAH12, AKAP9, DNAH8, VPS13B
450.0-452.0 kDa	2	0	0		routine	AKAP9, ABCA12
452.0-454.0 kDa	6	0	0		routine	NEB, ABCA12, SPTBN5, ABCA13, CMYA5, DNAH12
454.0-456.0 kDa	4	0	0		routine	AKAP9, CMYA5, AKAP9, FAT4
456.0-458.0 kDa	2	0	0		routine	AKAP9, DNAH8
458.0-460.0 kDa	1	0	0		routine	DNAH7
460.0-462.0 kDa	1	0	0		routine	ABCA13
462.0-464.0 kDa	1	0	0		routine	DNAH3
464.0-466.0 kDa	1	0	0		routine	FAT4
470.0-472.0 kDa	1	0	1		routine	PRKDC
472.0-474.0 kDa	2	0	2		routine	PRKDC, PRKDC
474.0-476.0 kDa	6	0	0	Actin family	routine	HECTD4, PKD1 [Actin family], ANK3, ANK3, LRP1B, ANK3
476.0-478.0 kDa	5	0	0	Actin family	routine	PKD1 [Actin family], DNAH8, PKD1 [Actin family], NEB, DNAH9
480.0-482.0 kDa	3	0	0	Actin family	routine	DNAH1, PKD1 [Actin family], DYNC2H1
482.0-484.0 kDa	1	0	0		routine	FAT3
484.0-486.0 kDa	3	0	0	Actin family	routine	FAT2, PKD1 [Actin family], DYNC2H1
486.0-488.0 kDa	5	0	0		routine	CMYA5, DYNC2H1, FAT3, LOC101748756, DYNC2H1
488.0-490.0 kDa	3	0	1		routine	FAT1, LOC101748756, PRKDC
490.0-492.0 kDa	5	0	0		routine	VPS13D, VPS13D, LOC101748756, VPS13D, LOC101748756
494.0-496.0 kDa	2	0	0		routine	STARD9, FAT2
496.0-498.0 kDa	1	0	0		routine	NEB
498.0-500.0 kDa	4	0	0		routine	DNAH17, PCNT, CMYA5, DNAH9
500.0-502.0 kDa	3	0	0		routine	STARD9, PCNT, BIRC6
502.0-504.0 kDa	1	0	0		routine	FAT3
504.0-506.0 kDa	4	0	0		routine	PCNT, FAT3, PCNT, FAT3
506.0-508.0 kDa	7	0	0		routine	DNAH17, BIRC6, LOC101748756, FAT3, LRP1, LRP1, MYCBP2
508.0-510.0 kDa	4	0	0		routine	FAT3, FAT1, MYCBP2, FAT1
510.0-512.0 kDa	3	0	0		routine	MYCBP2, FAT1, FAT1
512.0-514.0 kDa	2	0	0		routine	LOC101748756, MYCBP2
514.0-516.0 kDa	1	0	0		routine	FAT1
516.0-518.0 kDa	2	0	0		routine	DYNC1H1, MYCBP2
518.0-520.0 kDa	1	0	0		routine	MYCBP2
520.0-522.0 kDa	6	0	0		routine	SACS, SACS, SACS, DYNC1H1, LOC101748756, SACS
522.0-524.0 kDa	8	0	0		routine	LRP2, APOB, LRP2, APOB, SACS, DNAH10, APOB, LRP2
524.0-526.0 kDa	1	0	0		routine	HERC2
526.0-528.0 kDa	2	0	0		routine	DNAH10, LOC101748756
528.0-530.0 kDa	2	0	0		routine	HERC2, DNAH8
530.0-532.0 kDa	3	0	0		routine	DYNC1H1, DYNC1H1, DNAH5
532.0-534.0 kDa	7	0	0		routine	HERC1, DNAH5, HERC1, DYNC1H1, DNAH5L, DYNC1H1, HERC1
534.0-536.0 kDa	2	0	0		routine	DNAH5, HERC1
536.0-538.0 kDa	2	0	0		routine	DNAH8, DNAH8
540.0-542.0 kDa	2	0	0	Histone family	routine	KMT2C [Histone family], RYR3
542.0-544.0 kDa	1	0	0	Histone family	routine	KMT2C [Histone family]
544.0-546.0 kDa	1	0	0	Histone family	routine	KMT2C [Histone family]
546.0-548.0 kDa	1	0	0	Histone family	routine	KMT2C [Histone family]
548.0-550.0 kDa	1	0	0		routine	ADGRV1
550.0-552.0 kDa	1	0	0		routine	STARD9
552.0-554.0 kDa	4	0	0		routine	RYR3, RYR3, RYR3, USH2A
554.0-556.0 kDa	3	0	0		routine	BLTP1, BLTP1, RYR2
556.0-558.0 kDa	3	0	0		routine	BLTP1, USH2A, RYR2
558.0-560.0 kDa	4	0	0		routine	BLTP1, LOC415641, STARD9, SSPO
560.0-562.0 kDa	3	0	0		routine	SSPO, PCLO, PCLO
562.0-564.0 kDa	1	0	0		routine	PCLO
564.0-566.0 kDa	3	0	0		routine	RYR2, RYR2, RYR2
566.0-568.0 kDa	1	0	0		routine	HYDIN
568.0-570.0 kDa	1	0	0		routine	PCLO
570.0-572.0 kDa	1	0	0	Histone family	routine	KMT2D [Histone family]
572.0-574.0 kDa	4	0	0	Histone family	routine	KMT2D [Histone family], UBR4, KMT2D [Histone family], UBR4
576.0-578.0 kDa	2	0	0		routine	HMCN1, HMCN1
582.0-584.0 kDa	2	0	0		routine	RNF213, RNF213
584.0-586.0 kDa	1	0	0		routine	RNF213
586.0-588.0 kDa	2	0	0		routine	HMCN1, HMCN1
588.0-590.0 kDa	1	0	0	Actin family	routine	MACF1 [Actin family]
590.0-592.0 kDa	2	0	0		routine	HMCN1, RNF213
604.0-606.0 kDa	1	0	0		routine	HMCN1
612.0-614.0 kDa	1	0	0	Actin family	routine	MACF1 [Actin family]
614.0-616.0 kDa	2	0	0	Actin family	routine	MACF1 [Actin family], MACF1 [Actin family]
632.0-634.0 kDa	1	0	0		routine	MDN1
646.0-648.0 kDa	1	0	0		routine	DST
654.0-656.0 kDa	1	0	0		routine	ADGRV1
666.0-668.0 kDa	1	0	0		routine	DST
678.0-680.0 kDa	1	0	0		routine	DST

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
684.0-686.0 kDa	2	0	0		routine	ADGRV1, DST
700.0-702.0 kDa	1	0	0		routine	AHNAK2
706.0-708.0 kDa	1	0	0		routine	NEB
734.0-736.0 kDa	1	0	0		routine	NEB
736.0-738.0 kDa	1	0	0		routine	NEB
796.0-798.0 kDa	1	0	0		routine	SYNE2
800.0-802.0 kDa	1	0	0		routine	SYNE2
802.0-804.0 kDa	1	0	0		routine	OBSCN
804.0-806.0 kDa	2	0	0		routine	SYNE2, OBSCN
822.0-824.0 kDa	2	0	0	Actin family	routine	OBSCN, MACF1 [Actin family]
826.0-828.0 kDa	2	0	0	Actin family	routine	DST, MACF1 [Actin family]
828.0-830.0 kDa	2	0	0	Actin family	routine	MACF1 [Actin family], MACF1 [Actin family]
830.0-832.0 kDa	1	0	0	Actin family	routine	MACF1 [Actin family]
848.0-850.0 kDa	1	0	0		routine	OBSCN
866.0-868.0 kDa	2	0	0		routine	DST, DST
868.0-870.0 kDa	1	0	0		routine	OBSCN
870.0-872.0 kDa	1	0	0		routine	DST
878.0-880.0 kDa	1	0	0		routine	DST
880.0-882.0 kDa	1	0	0		routine	OBSCN
882.0-884.0 kDa	1	0	0		routine	DST
890.0-892.0 kDa	2	0	0		routine	OBSCN, OBSCN
900.0-902.0 kDa	1	0	0		routine	OBSCN
914.0-916.0 kDa	1	0	0		routine	OBSCN
3614.0-3616.0 kDa	1	0	0		routine	
3644.0-3646.0 kDa	2	0	0		routine	,
3654.0-3656.0 kDa	1	0	0		routine	
3684.0-3686.0 kDa	1	0	0		routine	
3704.0-3706.0 kDa	1	0	0		routine	
3714.0-3716.0 kDa	1	0	0		routine	
3820.0-3822.0 kDa	1	0	0		routine	
3840.0-3842.0 kDa	1	0	0		routine	
3894.0-3896.0 kDa	1	0	0		routine	

Proteins sorted from lowest to highest molecular weight

The PDF shows the first 120 proteins. The complete table is in proteome_wb_integrity_atlas.csv.

MW kDa	Gene	Accession	Protein	MW bin	Families	Audit note
0.64		P83308	FMRFamide-like neuropeptide	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.13		P19851	Neurokinin-A	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.23	AGT	P67885	Angiotensinogen (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.25	GNRH2	P68072	Gonadoliberin-2	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.27		P84553	Histone H1.A2 (Fragment)	0.0-2.0 kDa	Histone family	configured audit family match: Histone family
1.38		P19850	Substance P	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.47	ADA2	P84520	Adenosine deaminase 2 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.55		A0A8V0XA68	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.61	NTS	P13724	Neurotensin	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.84	ME1	Q92060	NADP-dependent malic enzyme (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
2.24	NPPC	P21805	C-type natriuretic peptide	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.69	MLN	Q9PRP6	Motilin	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.91	EIF5A2	A0A1L1S0Z7	Eukaryotic translation initiation factor 5A2	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.97		A0A8V1A5Z4	LASP1 neighbor	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.13	SCT	P01280	Secretin	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.19	AMPD1	P81073	AMP deaminase 1 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.21	GAL	P30802	Galanin	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.24	RPN1	P80896	Dolichyl-diphosphooligosaccharide--protein glycosyltransferase subunit 1 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.27		A0A8V1ADG6	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.38	HPX	P20057	Hemopexin (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.39		P20137	Glutathione S-transferase 5 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.48	CLPS	P11148	Colipase (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.73	UBLCP1	A0A1L1RUH3	Ubiquitin like domain containing CTD phosphatase 1	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.79	SLN	F1P0F7	Sarcolipin	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.82	BLACAT1	A0A8V1AD43	Bladder cancer associated transcript 1	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.93	RPN2	P80897	Dolichyl-diphosphooligosaccharide--protein glycosyltransferase subunit 2 (Fragment)	2.0-4.0 kDa		standard full-blot and antibody-validation checks
4.06	C12orf75	A0A452J810	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.08		P83145	Ribonuclease UK114 (Fragment)	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.13	CRABP2	P30370	Cellular retinoic acid-binding protein 2 (Fragment)	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.24	PYY	P29203	Peptide YY-like	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.26		A0A8V0ZFU7	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.34	USP49	F1NPW7	Ubiquitin specific peptidase 49	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.48		P80389	Antimicrobial peptide CHP1	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.50		A0A8V0X2M7	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.52		A0A3Q3AA92	Protein phosphatase 4 catalytic subunit	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.61	GLI2	A0A8V0Z2B0	GLI family zinc finger 2	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.66		A0A8V0XFW2	Senescence-associated protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.88		A0A8V0XVD9	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.91		A0A8V0XA39	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.99		A0A8V0X2W0	Cation-transporting P-type ATPase N-terminal domain-containing protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.01		A0A8V0XDA0	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.02		A0A8V1A172	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.24	TMSB4B	A0A3Q2UAP0	Thymosin beta	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.31		A0A8V0Z9D2	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.33	RPS20	A0A8V0XLC8	Ribosomal protein S20	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.33		A0A8V0ZBY8	TCL1 upstream neural differentiation-associated RNA	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.37		A0A8V1ABK8	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.43		A0A8V0YLA8	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.51	CHCHD7	A0A8V0XKX3	Coiled-coil-helix-coiled-coil-helix domain-containing protein 7	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.51		A0A8V0YTH9	WAP domain-containing protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.53	C7orf73	A0A1D5PNN4	Short transmembrane mitochondrial protein 1	4.0-6.0 kDa	Mitochondrial markers	configured audit family match: Mitochondrial markers
5.57		A0A8V0XA75	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.62	U2AF1	A0A8V0YBE5	U2 small nuclear RNA auxiliary factor 1	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.64		A0A8V0X4T1	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.68	COX14	A0A8V1A933	COX14, cytochrome c oxidase assembly factor	4.0-6.0 kDa	Mitochondrial markers	configured audit family match: Mitochondrial markers
5.70		A0A8V0XSF1	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.71	LOC100857579	A0A8V0X4T4	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.73		A0A8V0Y3E2	MICOS complex subunit MIC10	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.73		A0A8V0X886	KRAB domain-containing protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.77		A0A8V0ZMZ5	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.79	ATP5E	A0A3Q2TXM3	Uncharacterized protein	4.0-6.0 kDa	Mitochondrial markers	configured audit family match: Mitochondrial markers
5.81	PIGBOS1	A0A8V0ZSH1	PIGB opposite strand 1	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.81		A0A8V0YKP7	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.82	SPINK2	A0A8V0XH94	Serine peptidase inhibitor Kazal type 2	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.94	PMAIP1	A0A1D5PAR2	Phorbol-12-myristate-13-acetate-induced protein 1	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.96		A0A8V0ZKX6	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.97	DNAJB8	A0A8V1A959	DnaJ heat shock protein family (Hsp40) member B8	4.0-6.0 kDa	Heat shock proteins	configured audit family match: Heat shock proteins
5.98		A0A8V0Z7P7	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
6.03		P85000	Trypsin inhibitor CITI-1	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.04		P0C915	Overexpressed in colon carcinoma 1 protein homolog	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.05		A0A8V1AFL1	THAP-type domain-containing protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.09	PLN	P26677	Phospholamban	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.09	PLN	R4GK80	Phospholamban	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.11	TOMM5	A0A8V0X9V5	Translocase of outer mitochondrial membrane 5	6.0-8.0 kDa	Mitochondrial markers	configured audit family match: Mitochondrial markers
6.12		A0A8V0XPE6	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.12		A0A8V0WZT7	KRAB domain-containing protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.18		A0A8V1A686	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks

MW kDa	Gene	Accession	Protein	MW bin	Families	Audit note
6.21	COPS9	A0A3Q2UDI0	COP9 signalosome complex subunit 9	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.22		A0A8V0XV54	HCV F-transactivated protein 1	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.24	C14orf2	A0A3Q2UCU2	6.8 kDa mitochondrial proteolipid	6.0-8.0 kDa	Mitochondrial markers	configured audit family match: Mitochondrial markers
6.24	LOC107055098	A0A3Q2UFR1	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.24		A0A8V0X5W2	KRAB domain-containing protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.28	MT3	A4PBT0	Metallothionein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.29		A0A8V0ZGZ1	Leucine rich melanocyte differentiation associated	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.29		A0A8V0X3L5	KRAB domain-containing protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.31	KCTD9	A0A8V0ZF54	Potassium channel tetramerization domain containing 9	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.32	TOMM7	A0A1L1RYN1	Mitochondrial import receptor subunit TOM7 homolog	6.0-8.0 kDa	Mitochondrial markers	configured audit family match: Mitochondrial markers
6.32	UQCR11	E1BXT9	Ubiquinol-cytochrome c reductase, complex III subunit XI	6.0-8.0 kDa	Mitochondrial markers	configured audit family match: Mitochondrial markers
6.34	APELA	F1NXJ1	Apelin receptor early endogenous ligand	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.35	TNFRSF10B	A0A8V0ZA81	Tumor necrosis factor receptor superfamily, member 10b	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.37		A0A8V1A3S1	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.37		A0A8V0X5S1	KRAB domain-containing protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.38	MT-ATP8	P14093	ATP synthase F(0) complex subunit 8	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.38	BOLL	A0A8V0YT37	Boule homolog, RNA binding protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.38	RPL39	Q98TF5	Large ribosomal subunit protein eL39	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.41	SMIM3	A0A8V0Z3W9	Small integral membrane protein 3	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.41		A0A8V0ZP14	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.42	TRPC1	A0A8V0YZZ6	Transient receptor potential cation channel subfamily C member 1	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.45		A0A8V0YUW4	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.46		P68497	Metallothionein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.48		A0A8V0Z473	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.48		A0A8V0Z9K5	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.49		P40667	SRY-related protein CH3 (Fragment)	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.50		A0A8V0ZP36	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.50	LOC107051594	A0A3Q2U1J3	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.52	LSM5	A0A8V0ZB54	LSM5 homolog, U6 small nuclear RNA and mRNA degradation associated	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.52		P40666	SRY-related protein CH2 (Fragment)	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.53		P40669	SRY-related protein CH7 (Fragment)	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.53	ATP5E	A0A8V0YNP2	ATP5E synthase	6.0-8.0 kDa	Mitochondrial markers	configured audit family match: Mitochondrial markers
6.55		P40670	SRY-related protein CH31 (Fragment)	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.55		P40668	SRY-related protein CH4 (Fragment)	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.56		A0A8V0X8T5	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.56		P40671	SRY-related protein CH32 (Fragment)	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.57		P40665	SRY-related protein CH1 (Fragment)	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.58	ENHO	A0A1D5NVP7	Energy homeostasis associated	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.58		A0A8V0X5X3	KRAB domain-containing protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.60	GAL14	Q0E4V3	Gallinacin-14	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.60	CMC1	A0A8V0XHG3	COX assembly mitochondrial protein	6.0-8.0 kDa	Mitochondrial markers	configured audit family match: Mitochondrial markers
6.61		A0A8V1ADN4	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.61		A0A8V0XRF8	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks

Reported western-blot malpractice cases mapped to MW bins

These are reported public cases supplied in the malpractice input TSV. They are evidence examples for integrity-audit results sections, not substitution recommendations.

Case	Claim/protein	MW bin	Reported issue	Results sentence
ORI_Eckert_JBC2014_MEK3	MEK3 western blot	38.0-40.0 kDa	ORI reported MEK3 bands were compiled from a source image representing p44 expression in an unrelated experiment.	A public ORI case documents a human-cell western blot malpractice example in the 38-40 kDa bin, where MEK3-labeled bands were compiled from an unrelated p44 source image.
ORI_Eckert_JBC2014_p38d	p38δ western blot	40.0-42.0 kDa	ORI reported p38δ bands were compiled from a source image representing β-actin expression in an unrelated experiment.	The 40-42 kDa bin is supported by a public ORI case in which p38δ-labeled bands were compiled from a β-actin source image.
ORI_Eckert_PLoS2012_TAM67	TAM67-FLAG western blot	48.0-50.0 kDa	ORI reported TAM67-FLAG bands were generated using unrelated bands from a source image representing Cyclin A expression.	A public ORI case maps a relabeled western blot problem to the 48-50 kDa neighborhood, involving Cyclin A source material used for a TAM67-FLAG claim.
ORI_Yang_GAPDH_NFYA	GAPDH loading-control bands	36.0-38.0 kDa	ORI reported claimed GAPDH loading-control bands were actually a prominent nonspecific/background band from an NF-YA western blot.	The 36-38 kDa neighborhood is supported by an ORI-described case where a nonspecific NF-YA blot band was claimed as GAPDH.
CancerGeneTherapy_Liu2021_CTBP2_GAPDH	CTBP2 and GAPDH western blots	48.0-50.0 kDa	Retraction note reported apparent similarity in western blots presented for different proteins, including CTBP2 and GAPDH.	A retraction notice maps an apparent different-protein western blot similarity to the CTBP2/GAPDH region, spanning roughly the 36-38 and 48-50 kDa neighborhoods.
CancerGeneTherapy_Liu2021_CyclinD1_CDK4	Cyclin D1 and CDK4 western blots	34.0-36.0 kDa	Retraction note reported apparent similarity in western blots presented for Cyclin D1 and CDK4.	A retraction notice maps an apparent different-protein western blot similarity to the 32-36 kDa cell-cycle neighborhood involving Cyclin D1 and CDK4.
Das_UConn_Garlic_AKT	Akt / p-Akt western blots in Figure 6	54.0-56.0 kDa	UConn/ACS retraction materials reported data fabrication in Figures 6, 7, and 8; exact lane-level source images are not public.	The 54-56 kDa signaling region is supported as an audit-priority bin by the garlic-paper retraction, where Akt/p-Akt figure panels were within figures reported to contain fabricated data.
Das_UConn_Garlic_FOXO1	FoxO1 / p-FoxO1 western blots in Figure 6	70.0-72.0 kDa	UConn/ACS retraction materials reported data fabrication in Figures 6, 7, and 8; exact lane-level source images are not public.	The 70 kDa transcription-factor signaling region can be referenced as an audit-priority bin because FoxO1 panels occurred in a figure reported to contain fabricated data.
Das_UConn_Garlic_NFKB_PPAR_GLUT4	Nrf2, NF-kB p65, PPARα/δ, Glut-4 western blots in Figures 7 and 8	52.0-54.0 kDa	UConn/ACS retraction materials reported data fabrication in Figures 6, 7, and 8; exact lane-level source images are not public.	The 50-56 kDa metabolic/signaling region is an audit-priority region in the garlic-paper case because PPAR and Glut-4 panels were in figures reported to contain fabricated data.
Ilangovan_HSP25_HSP27	Hsp25/Hsp27-family western blots	22.0-24.0 kDa	Public investigation materials described Hsp-family western blot relabeling and unreliable/falsified blot images.	The 22-24 kDa small heat-shock-protein neighborhood can be cited as an audit-risk bin based on public investigation materials describing Hsp-family western blot relabeling.

Integrity-audit checklist

- Ask for the full, uncropped blot with ladder and lane labels. Cropped panels are not enough.
- Confirm expected molecular weight and apparent molecular weight. Do not treat molecular-weight agreement as proof of identity.
- For configured marker proteins, request source-imager metadata and original antibody-validation records.
- Check whether a marker band is near common markers such as GAPDH, actin, tubulin, HSPs, lamins, VDAC, PCNA, or other configured controls.
- For signaling proteins, verify phospho/total controls, controls for activation state, and an independent identity control when central to the claim.
- In crowded or high-audit bins, prioritize raw-data provenance, full blot context, and orthogonal validation.
- Interpret mapped malpractice cases only as documented audit precedents. They do not imply that proteins in the same bin are interchangeable.

Generated files

- proteome_wb_integrity_atlas.csv - complete per-protein atlas sorted by theoretical MW.
- mw_bin_summary.csv - complete molecular-weight-bin audit summary.
- reported_malpractice_bins.csv - generated when a malpractice TSV is supplied.
- plots/ - four standalone PNG figures embedded in this report.
- western_blot_integrity_atlas_full_report_with_plots.pdf - this consolidated report.
- western_blot_integrity_atlas_report.pdf - byte-identical compatibility copy of the consolidated report.

Input configuration

Input	File
Proteome FASTA	chicken_UP000000539.fasta
Marker file	chicken_marker_genes.tsv
Signaling-prefix file	chicken_signaling_prefixes.tsv
Family-pattern file	chicken_family_patterns.tsv
Species/report label	Gallus gallus